

ASEN 5090 Lecture 10

Multipath



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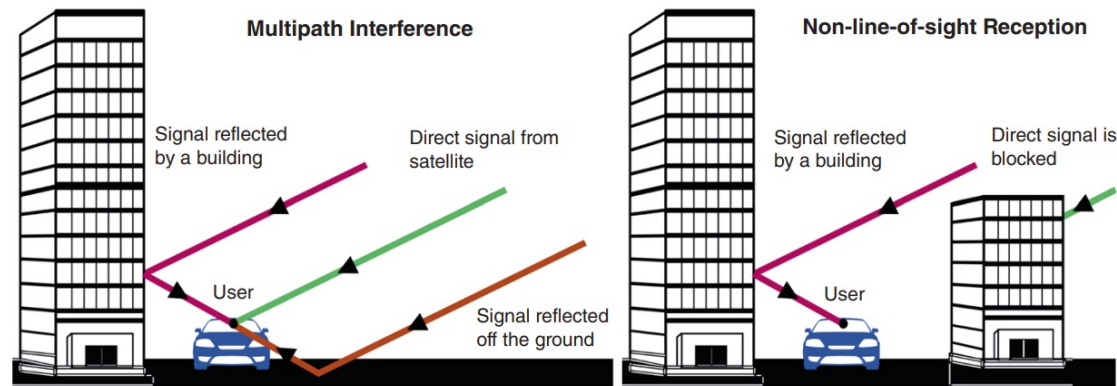
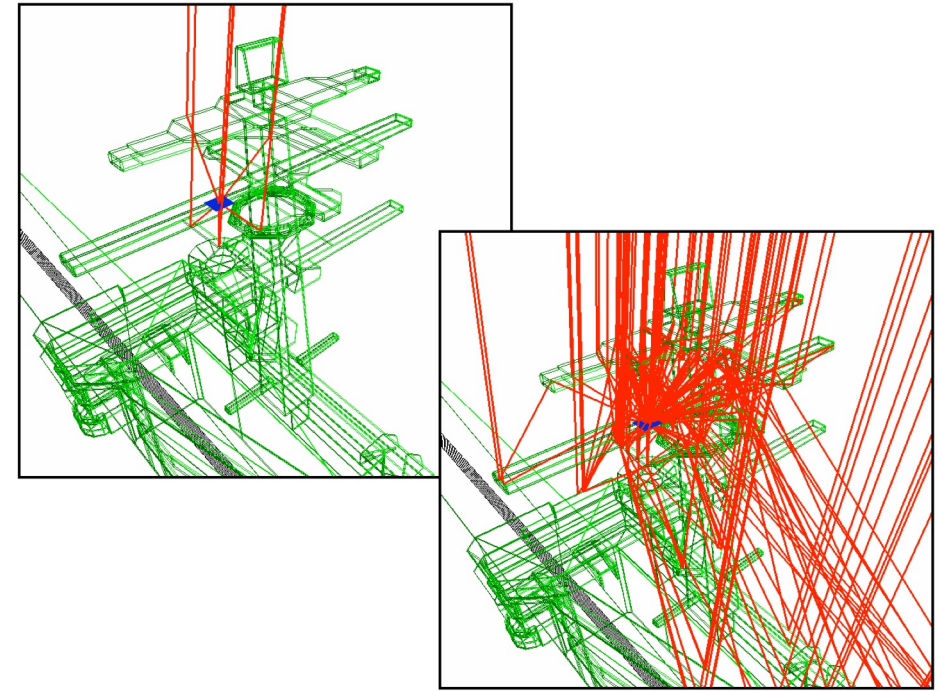


Figure 22.1 Multipath interference and NLOS reception.

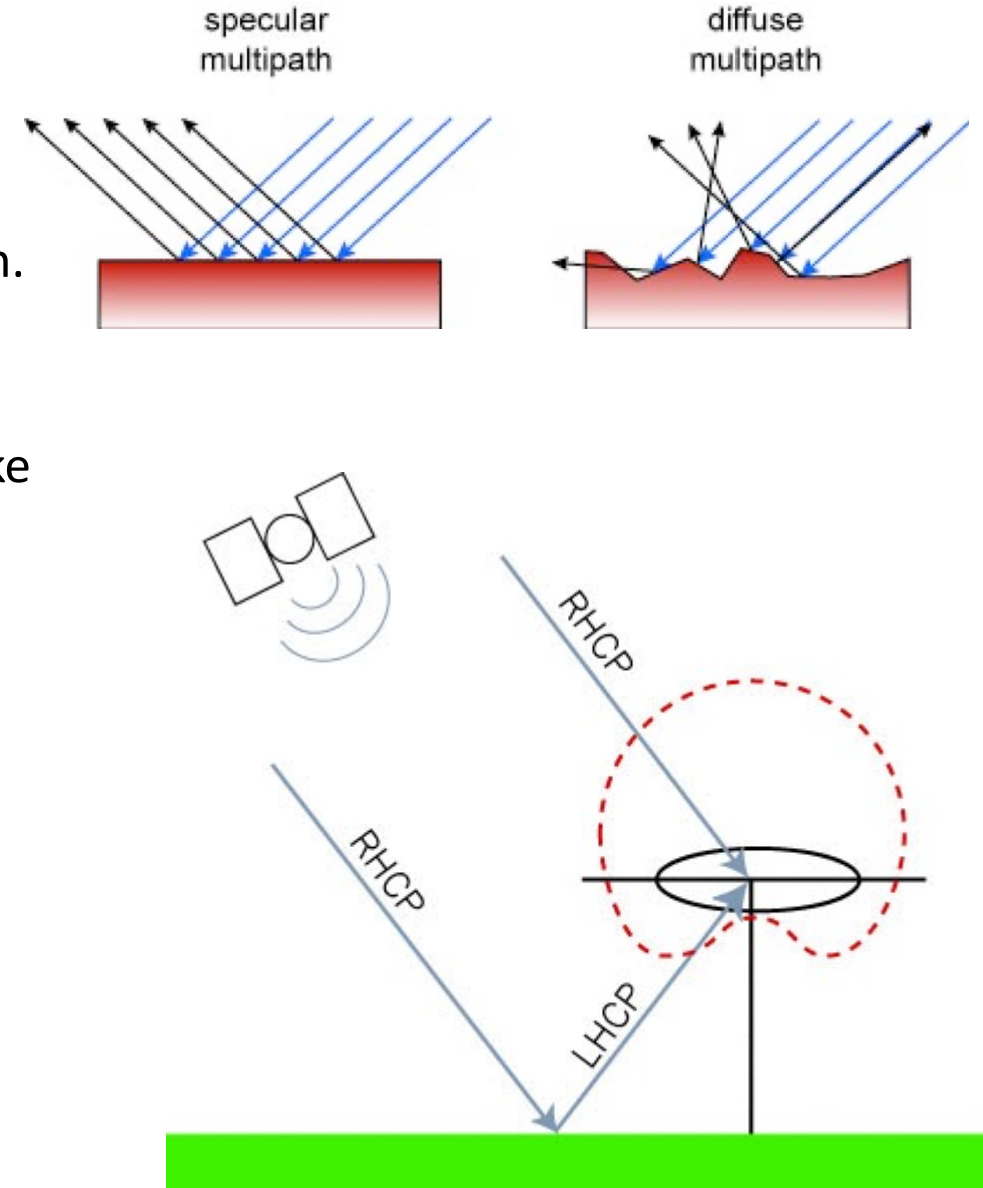
Outline

- What is multipath?
- How can you avoid it?
- How does multipath affect GPS pseudorange measurements?
carrier phase & SNR data?
- How can the effects of multipath on measurements be observed?
- How can the effects of multipath be mitigated?
- Can multipath be useful?



Multipath

- Definition: A signal that arrives at the antenna by an indirect path.
- Multipath is always delayed relative to the direct signal.
- **Diffuse** multipath results from rough surface reflections - noiselike
- **Specular** multipath results from smooth surfaces – causes oscillations in pseudorange, phase, SNR data
- **Specular reflection** is the mirror-like reflection of light (or other waves) from a surface, in which light from an incoming direction (a ray) is reflected into a single outgoing direction.



Let's talk a little more about this picture

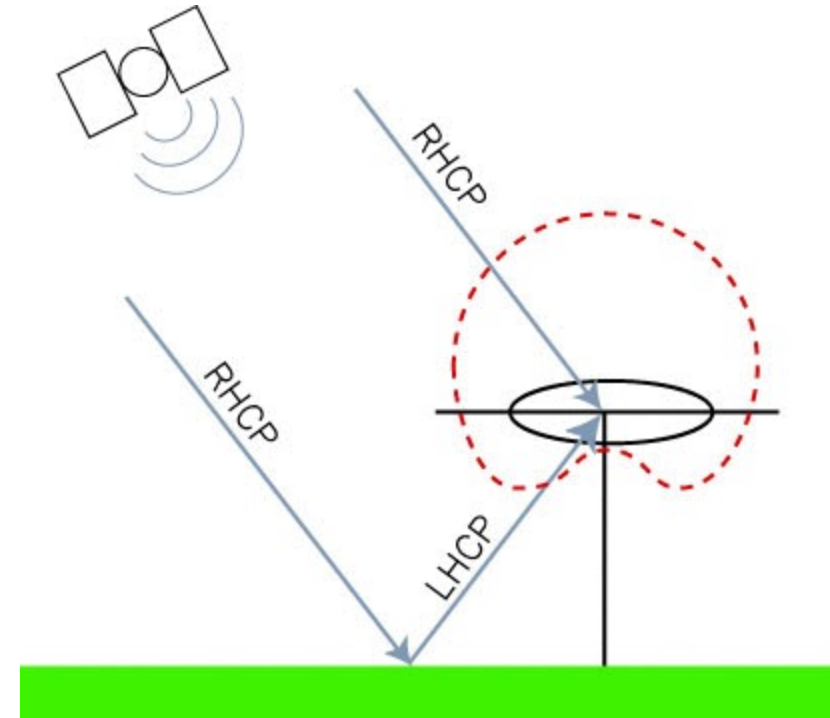


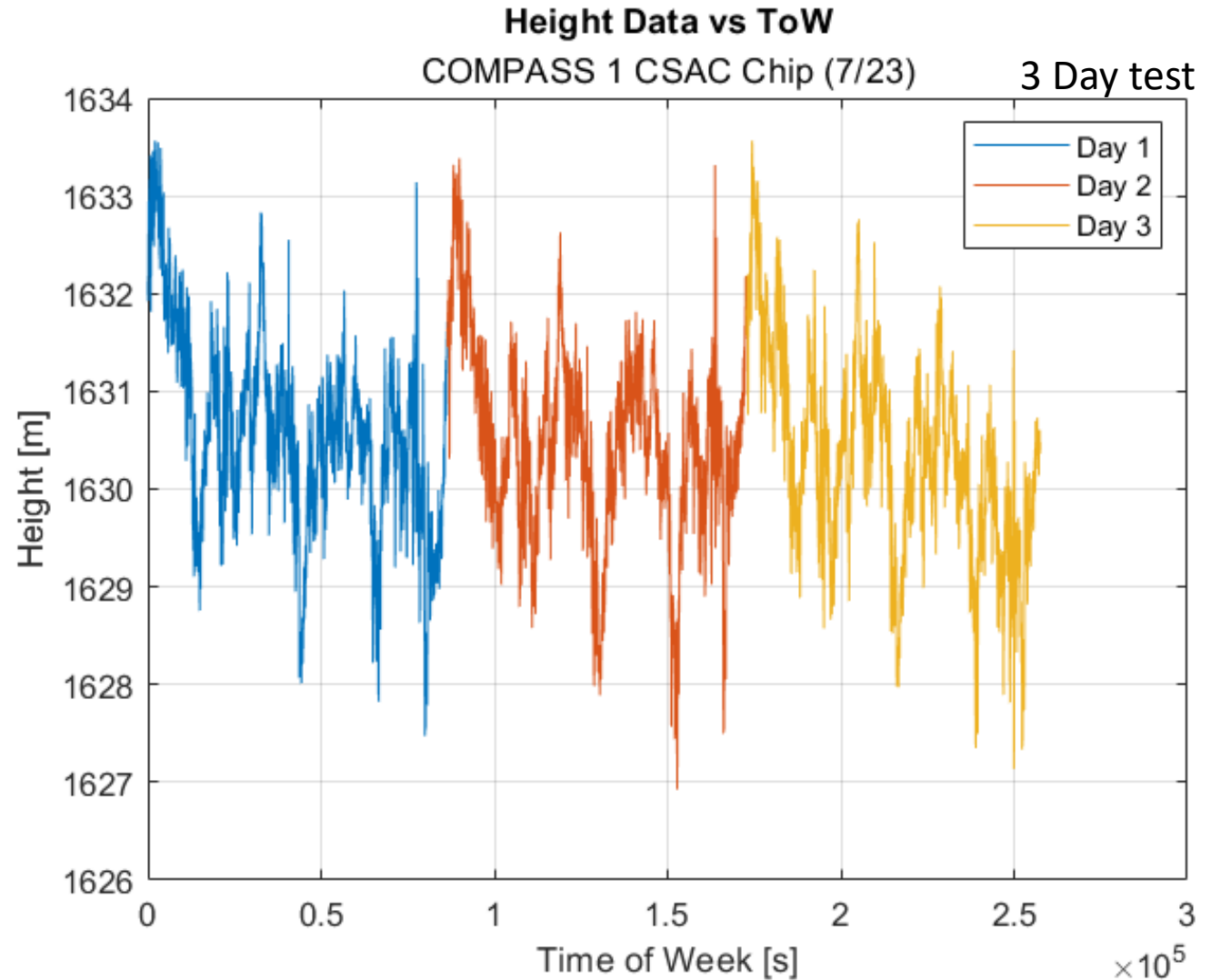
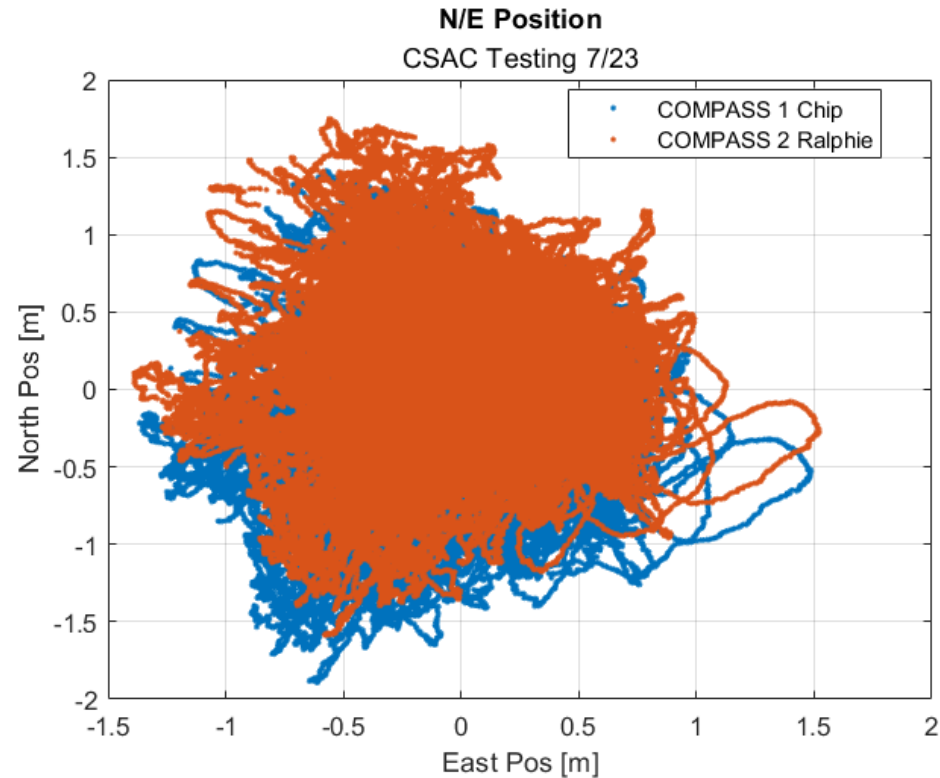
Photo of COMPASS Lab Antenna on AERO roof



Photo credit Harrison Bourne

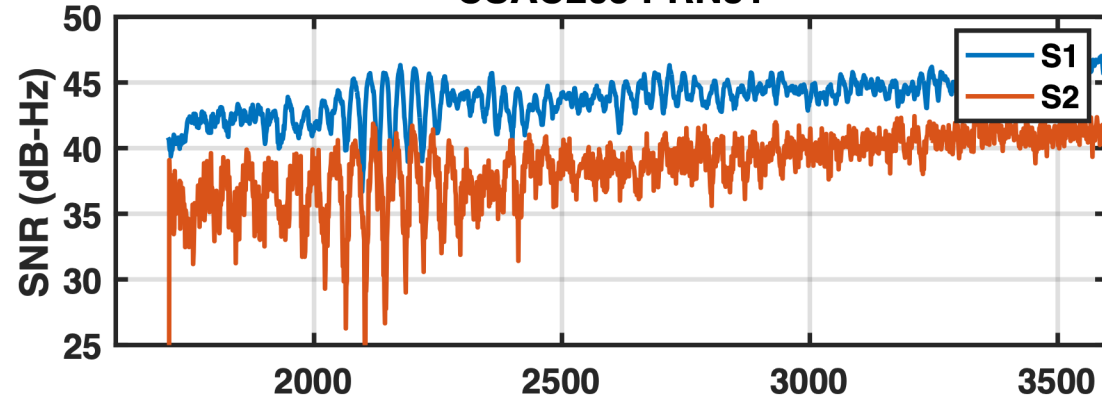


Example of North-East and Height Solutions

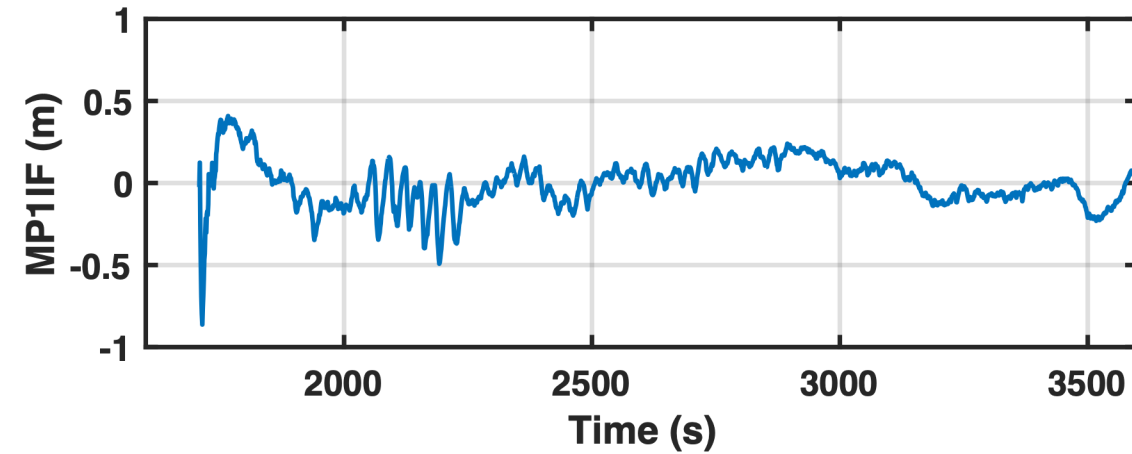


Multipath Effects on Individual Satellite SNR & Pseudorange

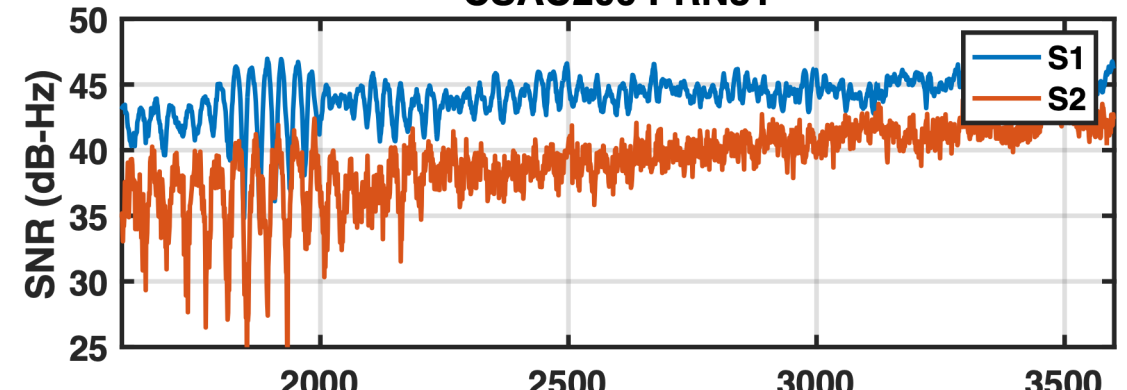
CSAC205 PRN31



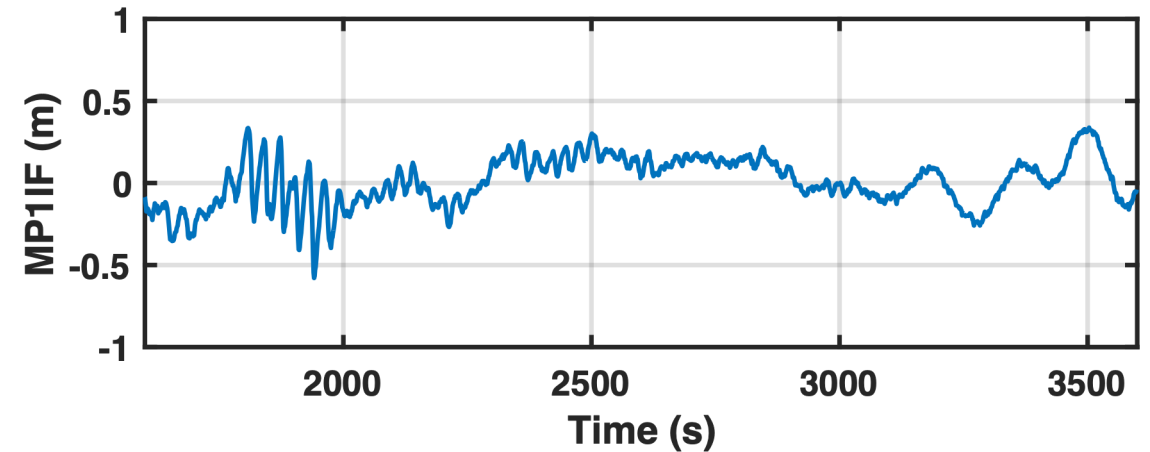
Time (s)



CSAC206 PRN31

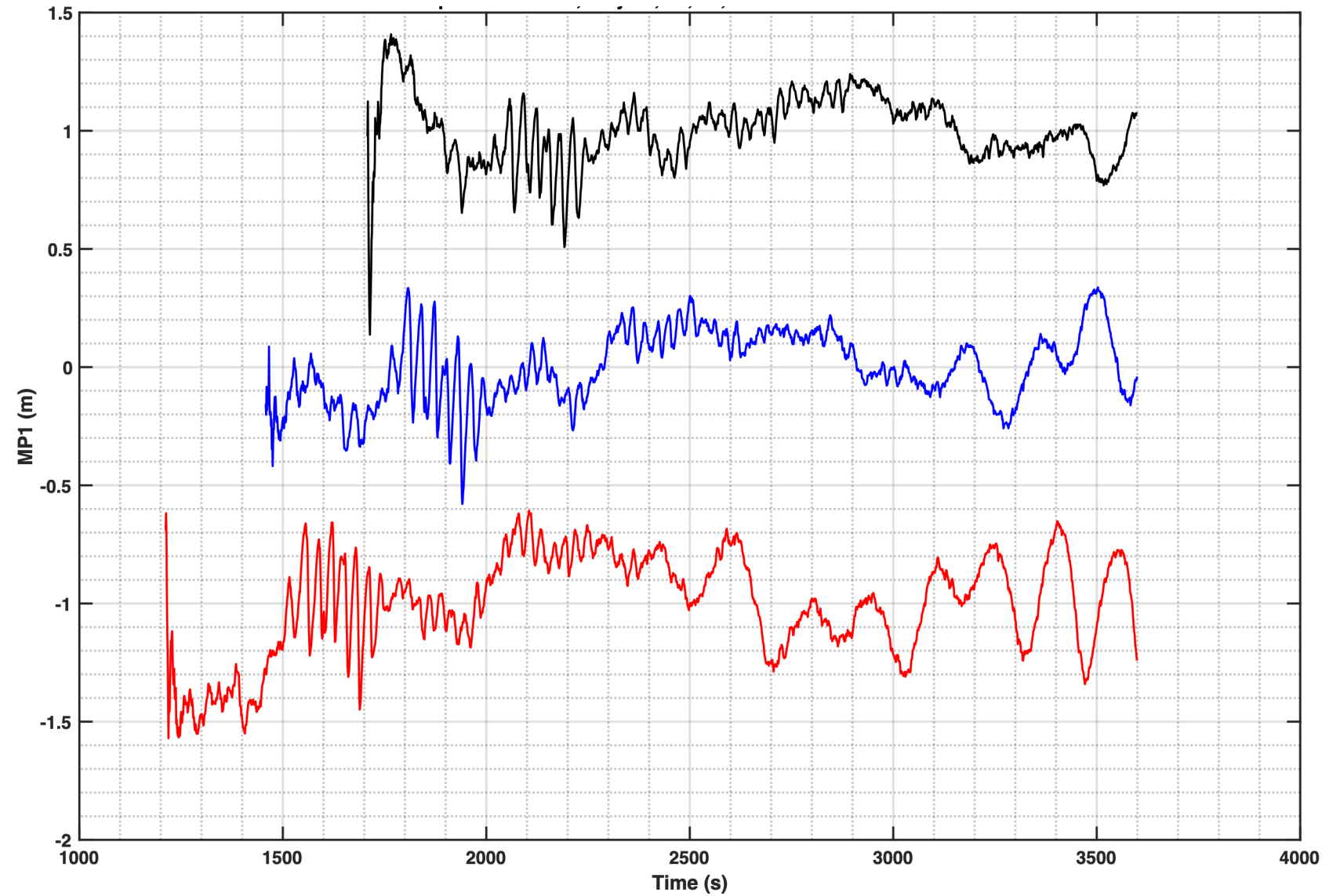


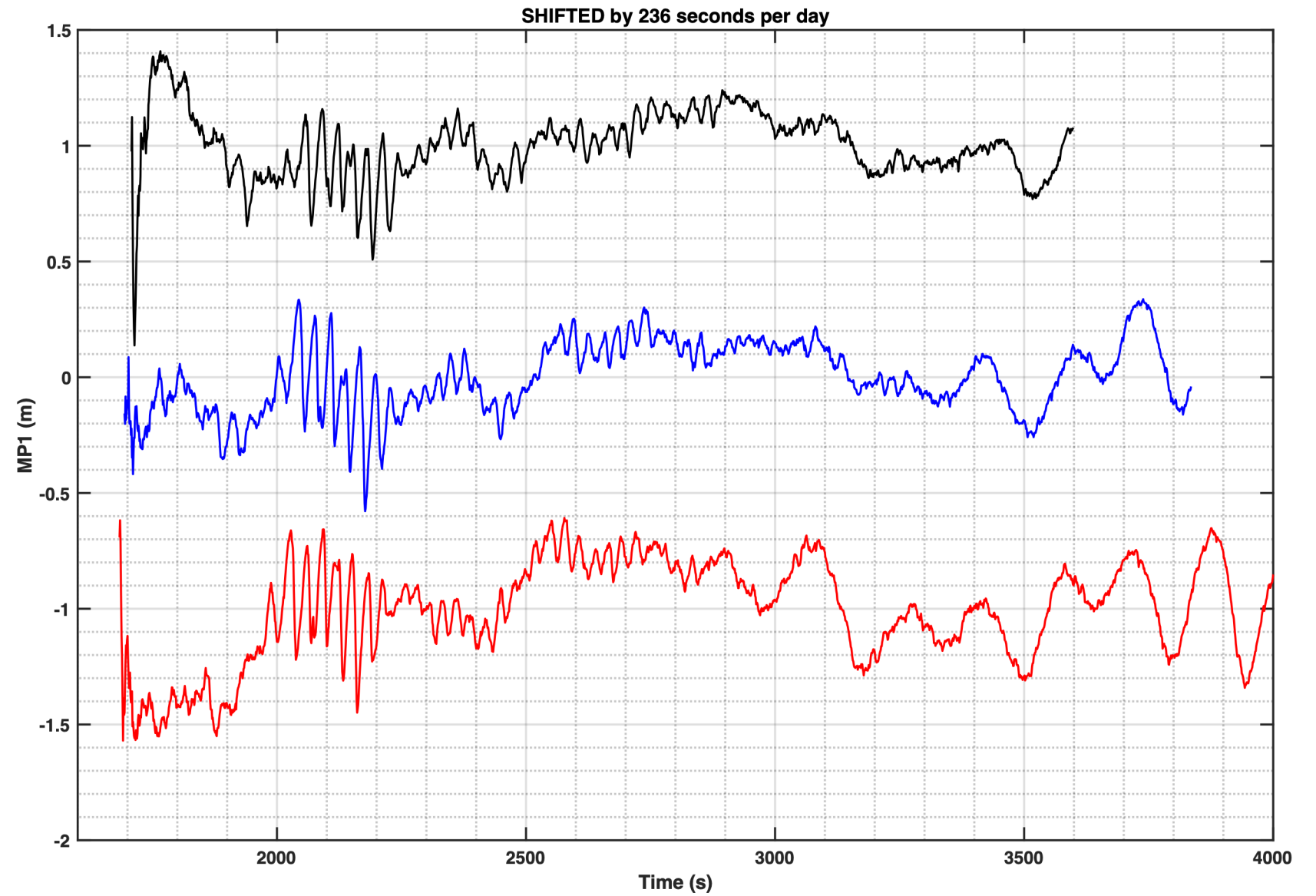
Time (s)



MULTIPATH OBSERVABLE for PRN31, July 23, 24, 25, 2021 – COMPASS LAB Roof Antenna

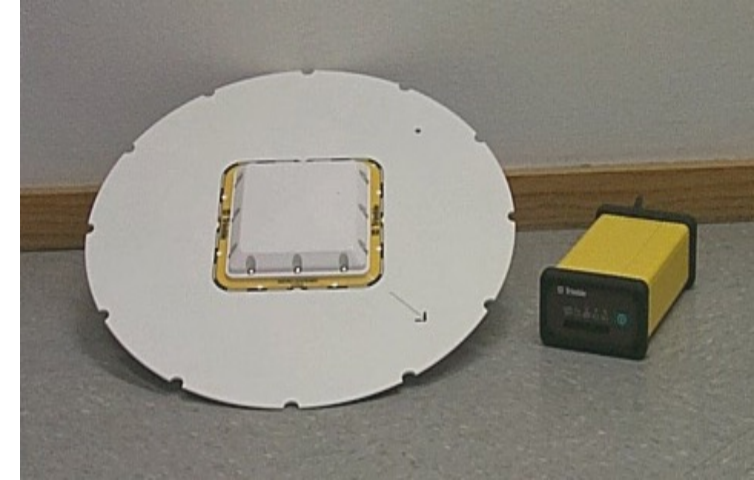
These are pseudorange errors due to multipath reflections





Avoiding and reducing multipath

- Antenna based mitigation
 - RHCP vs LHCP
 - Ground planes and choke rings
- Siting - install GPS sites away from reflecting materials
- Improved receiver technology
 - Narrow correlator spacing (will defer discussion until receiver section)
- Signal and data processing
 - Smoothing
 - Correlation in time

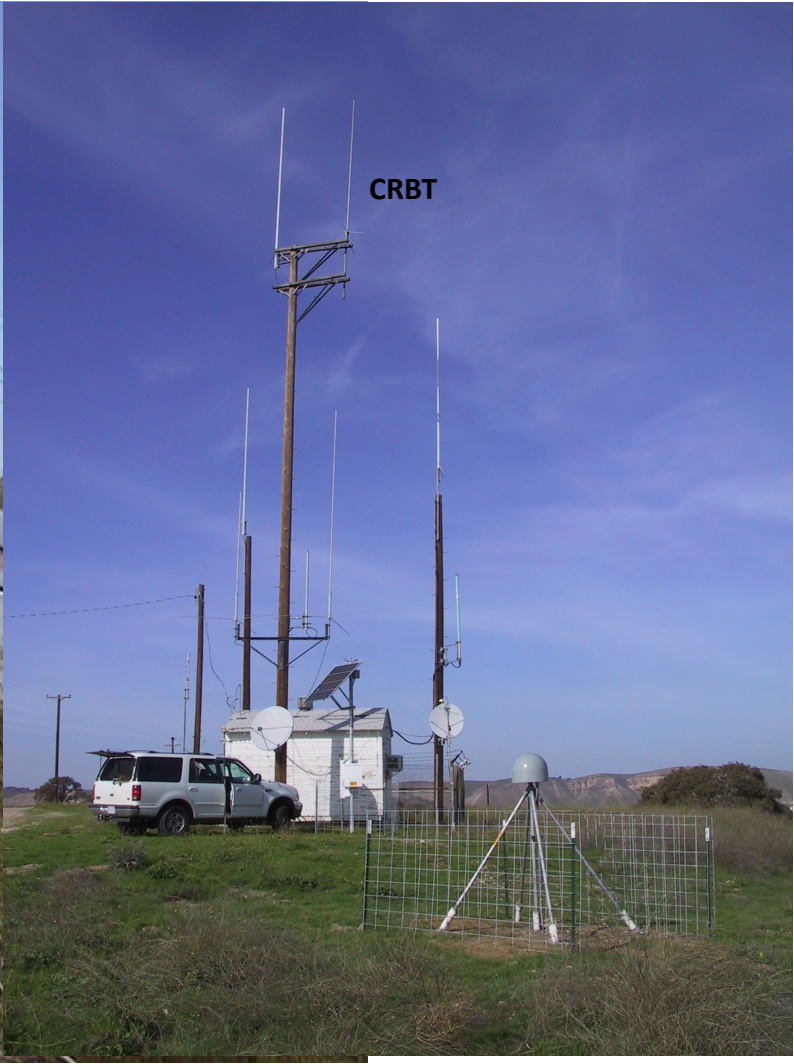
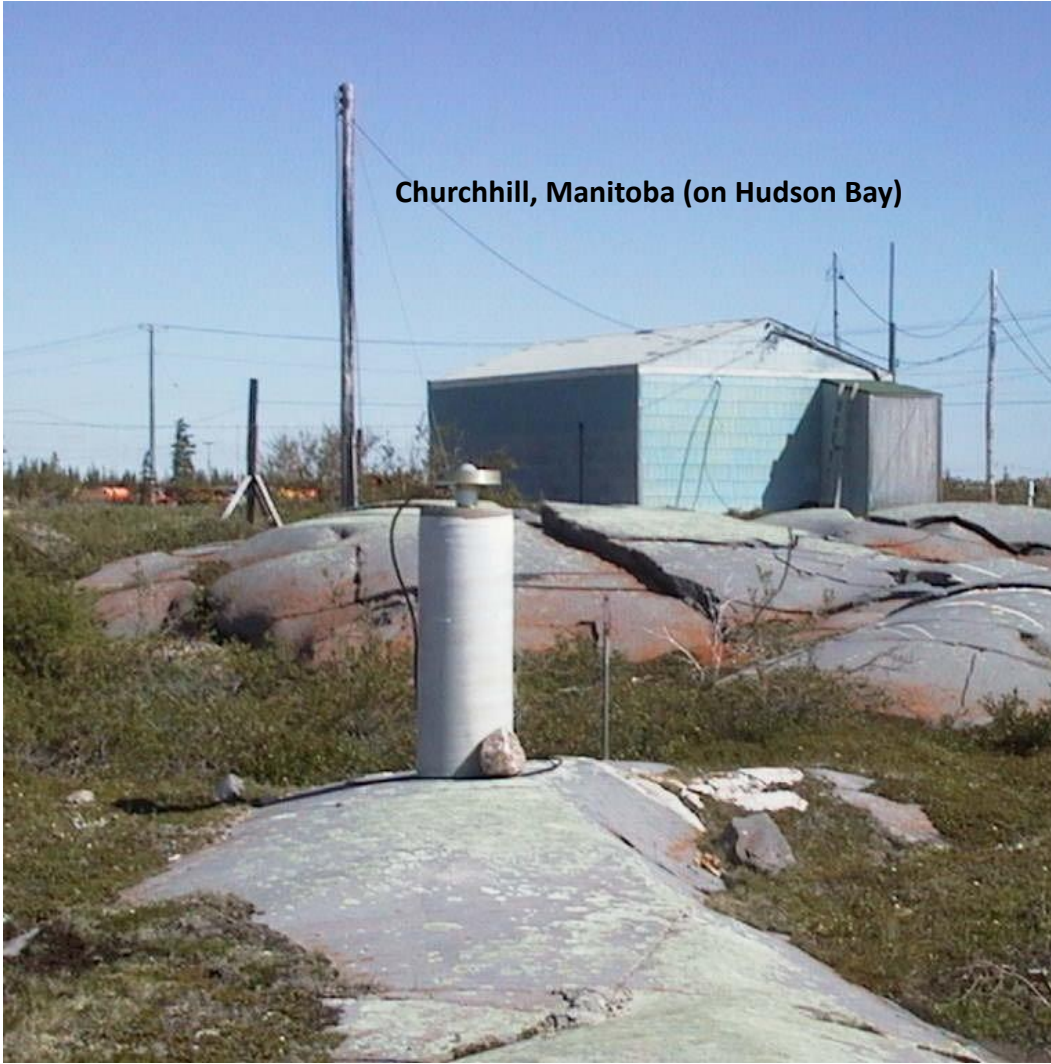


Antenna with ground plane



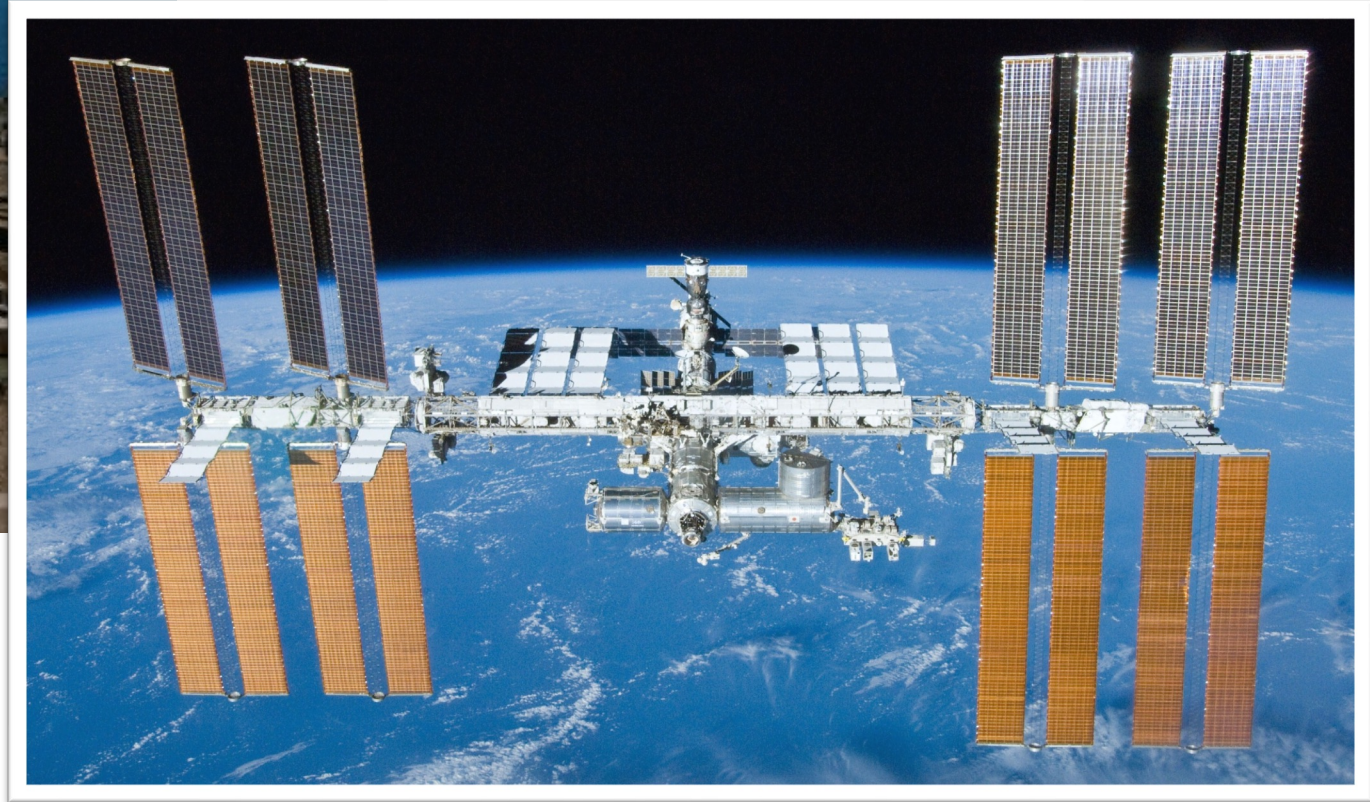
Choke ring antenna

Siting Examples

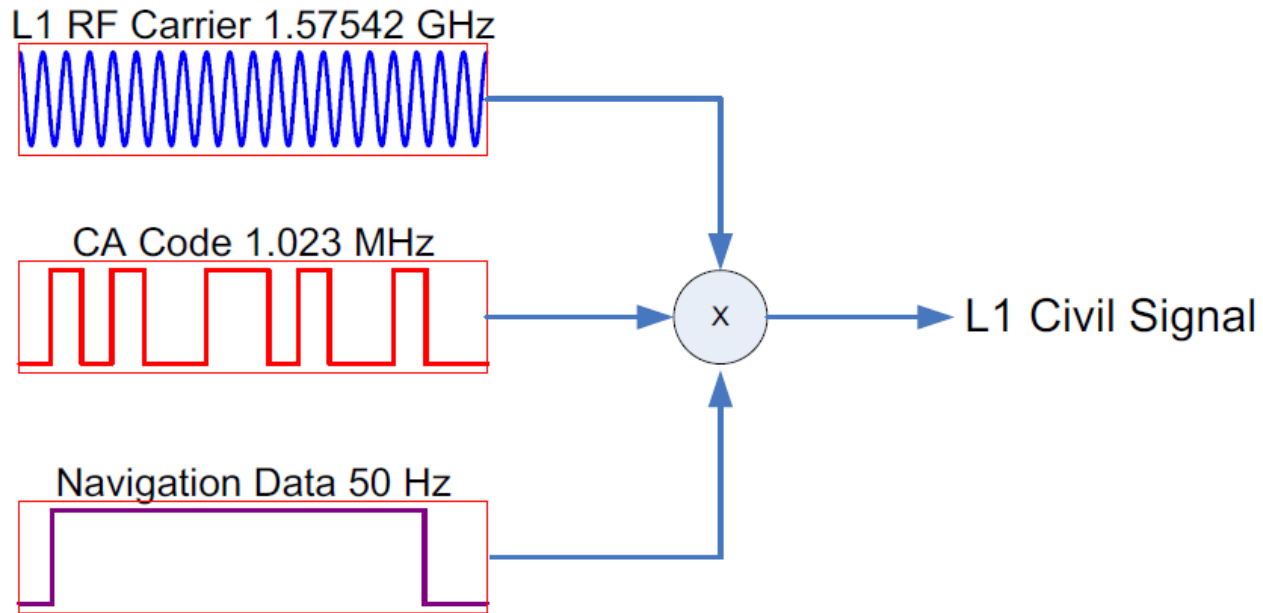


USS Eisenhower)

Not the best place to put a GPS antenna...

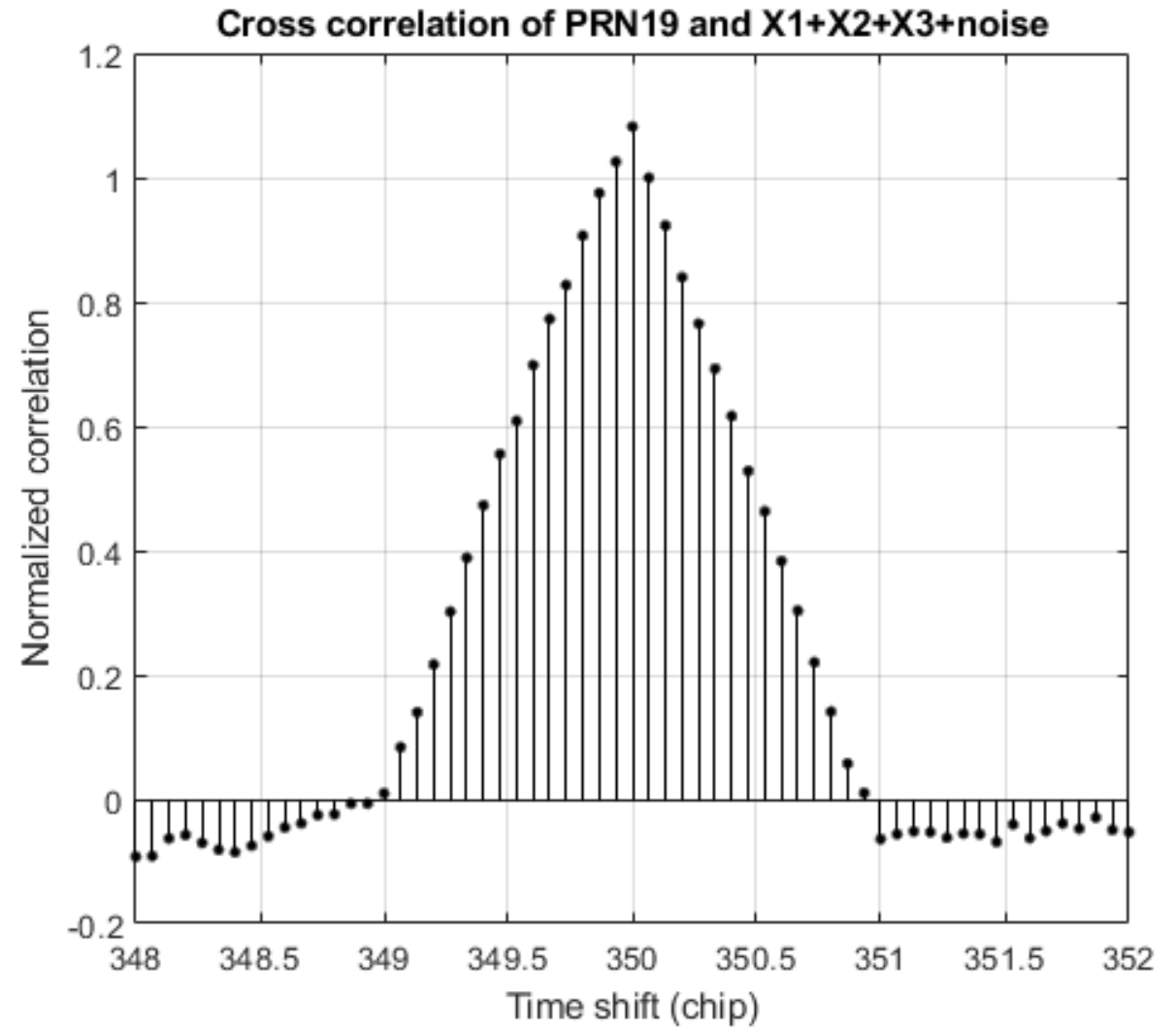
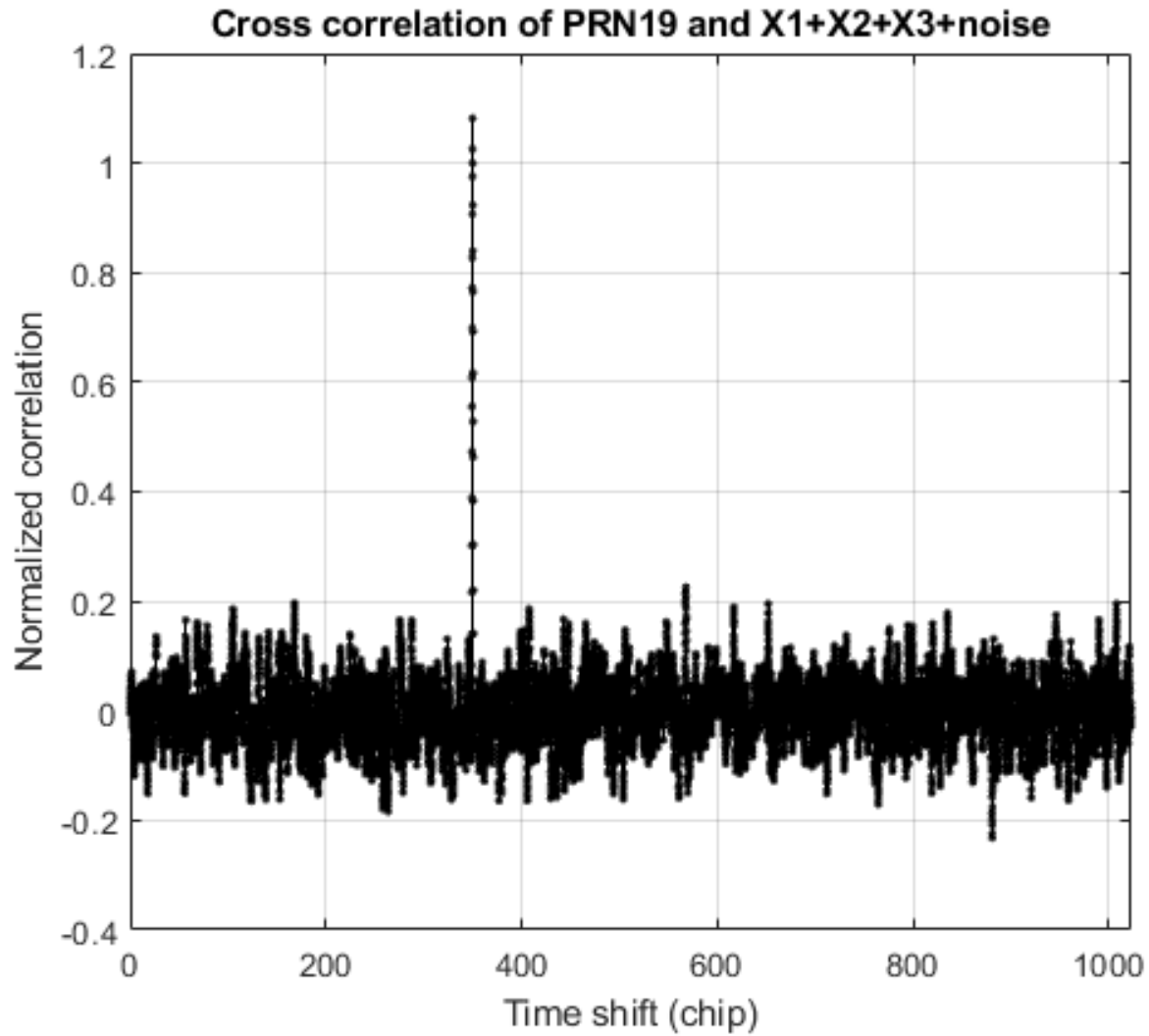


Review: GPS Signal Components



$$s(t) = \sqrt{C} D(t - \tau) c(t - \tau) \cos(2\pi f t + \varphi)$$

Recall: PRN code auto-correlation



Multipath Interference vs NLOS Reception

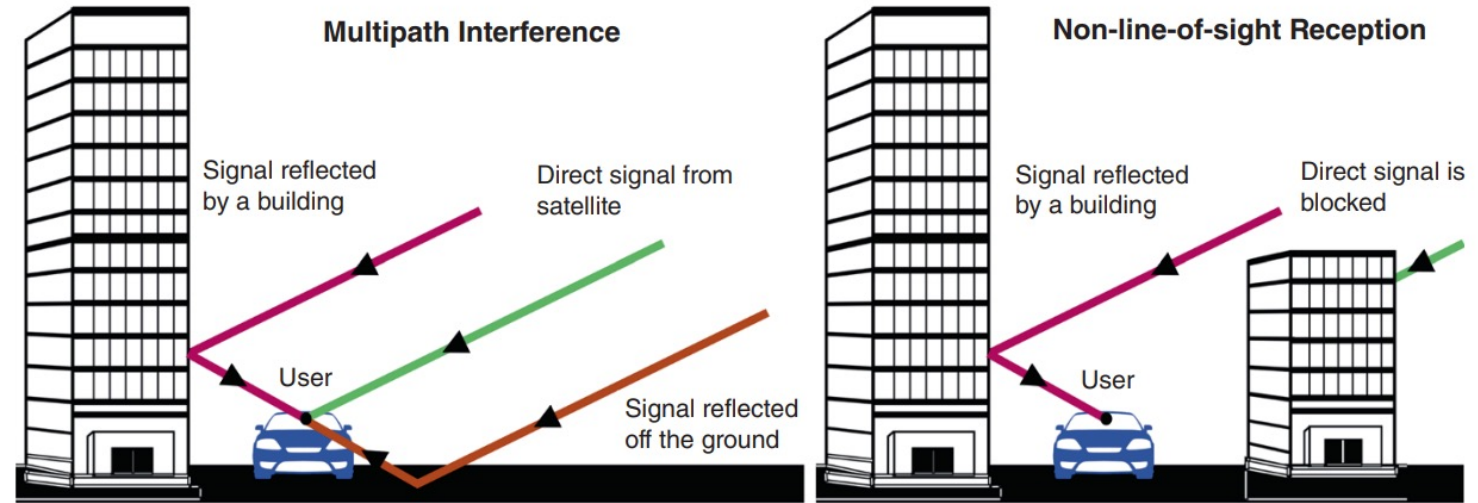
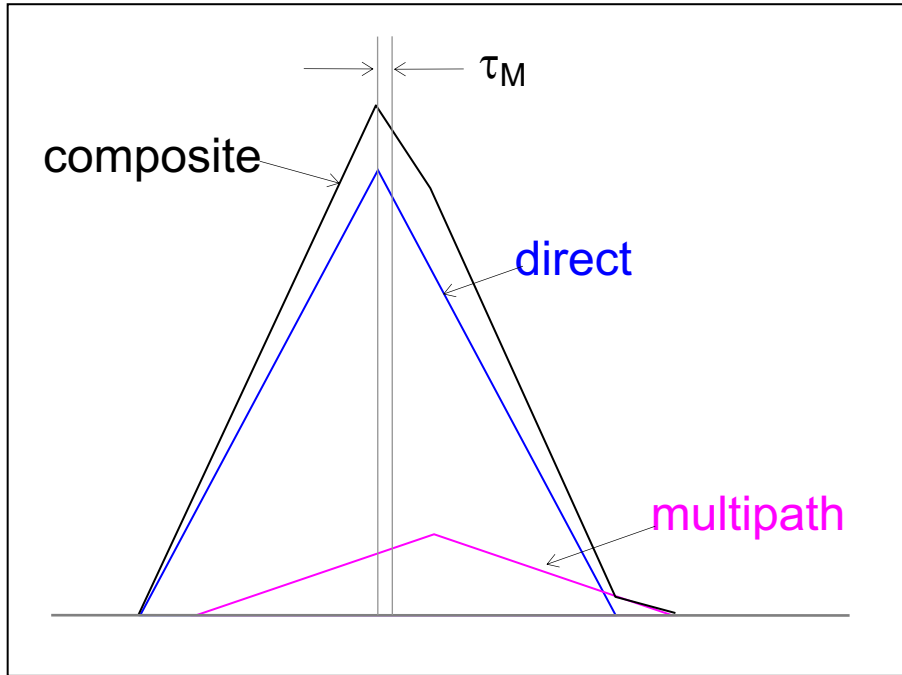


Figure 22.1 Multipath interference and NLOS reception.

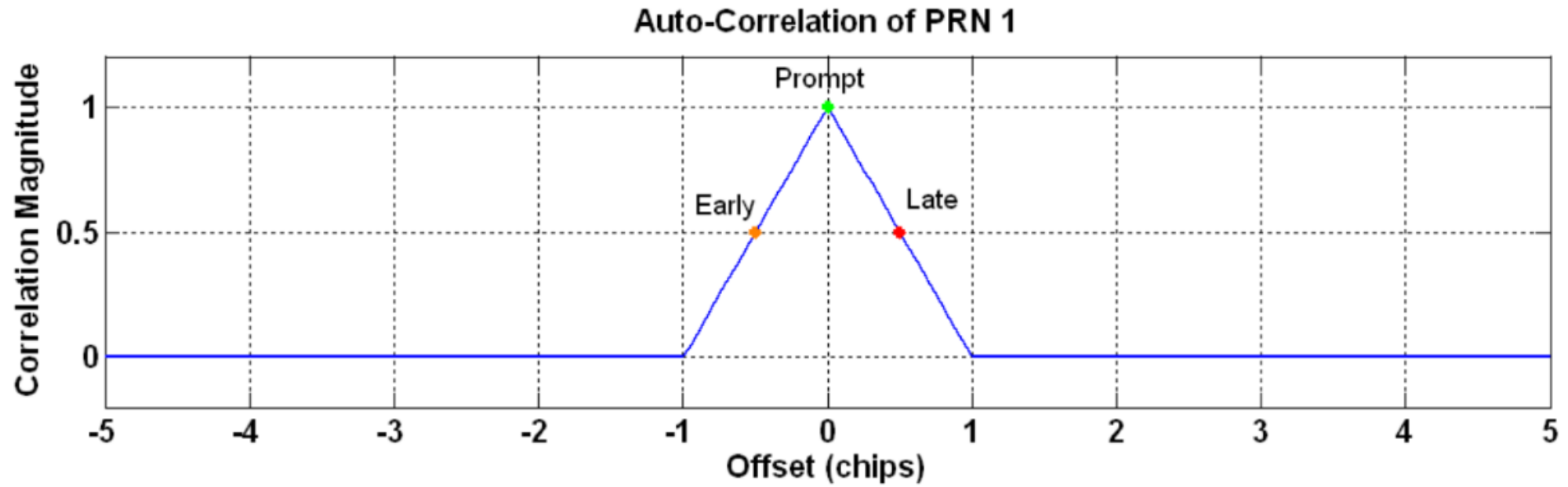
(21st Century PNT)

Multipath interference: the reflected signals distort the code correlation peak within the receiver.

NLOS reception occurs where the direct LOS signal is blocked, and the signal is received only via reflection. (usually 10s of meters, up to km)

Tracking the Peak

- Acquisition passes peak information to tracking
- In tracking, the goal is to know where the peak is – this relates to start of C/A code.
- Typically calculate three different correlation values

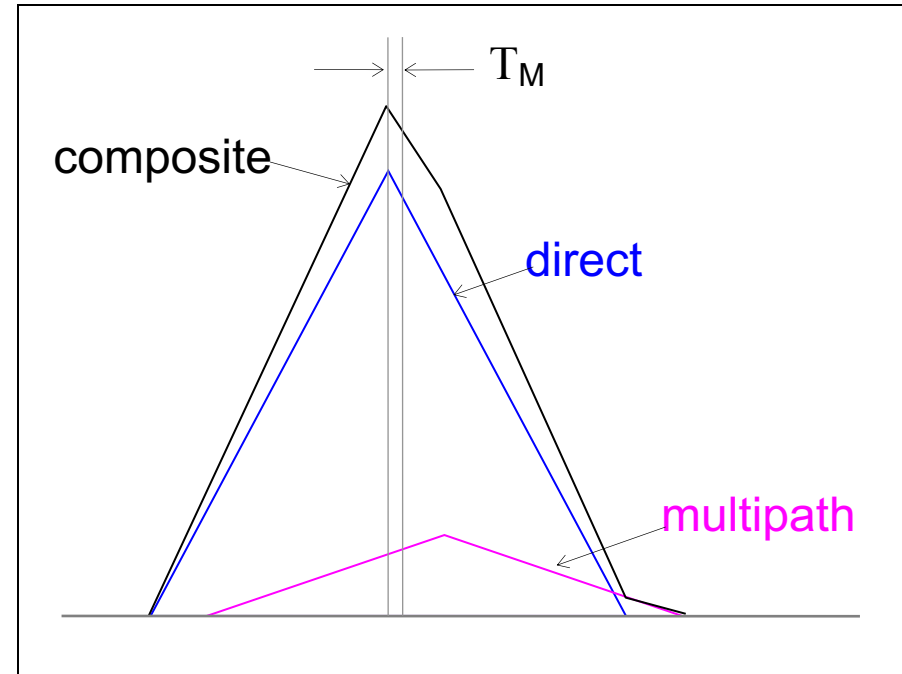


Effect of multipath on code/pseudorange measurement

$$\begin{aligned} & \left[R(\tau_M + T_{E/L}) - R(\tau_M - T_{E/L}) \right] \\ & + \alpha \left[R(\tau_M + T_{E/L} - \delta) - R(\tau_M - T_{E/L} - \delta) \right] \cos \psi = 0 \end{aligned}$$

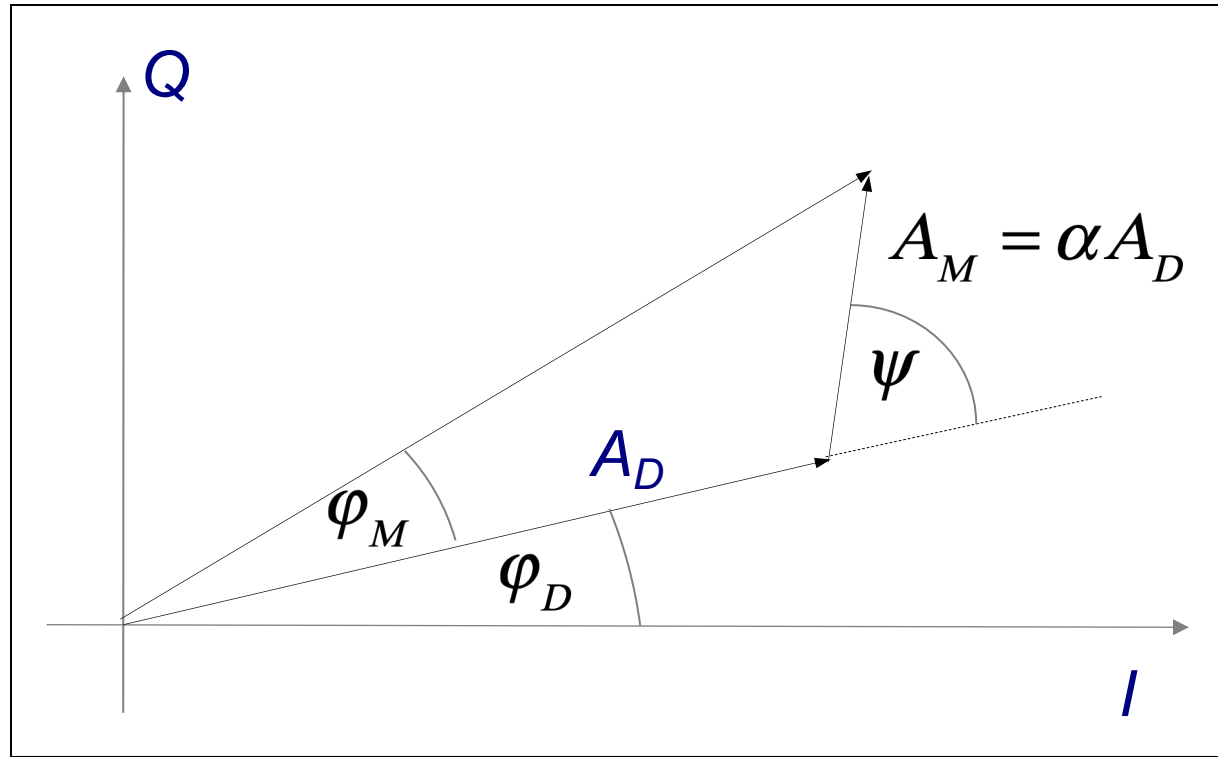
SHORT DELAY Approximation

$$\tau_M = \frac{\alpha \delta \cos \psi}{1 + \alpha \cos \psi}$$



Effect of multipath on phase measurements

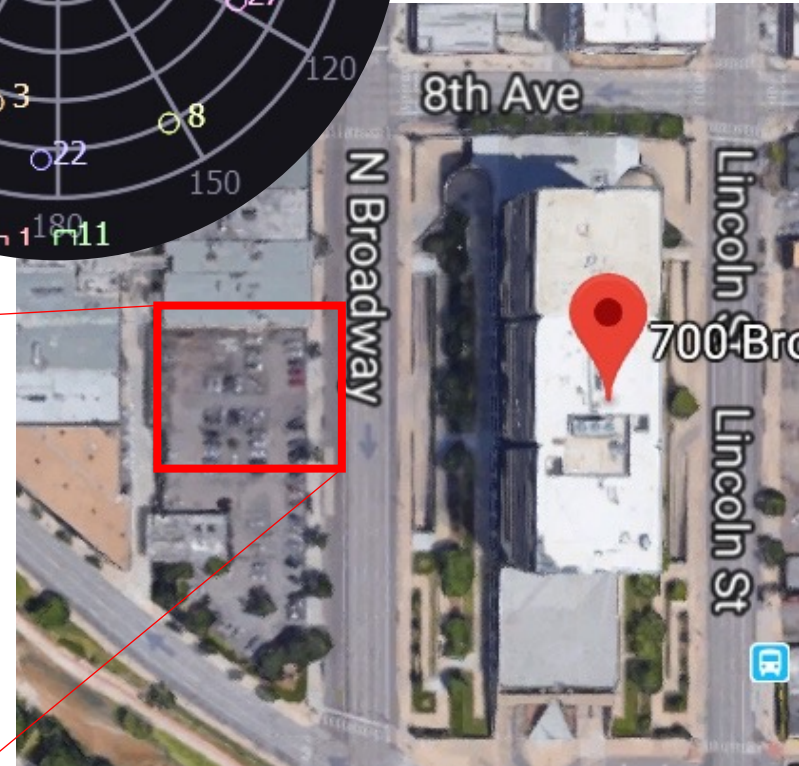
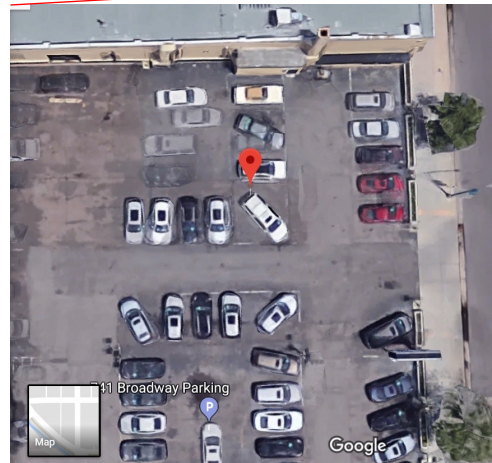
$$\tan \varphi_M = \frac{\alpha \sin \psi}{1 + \alpha \cos \psi}$$



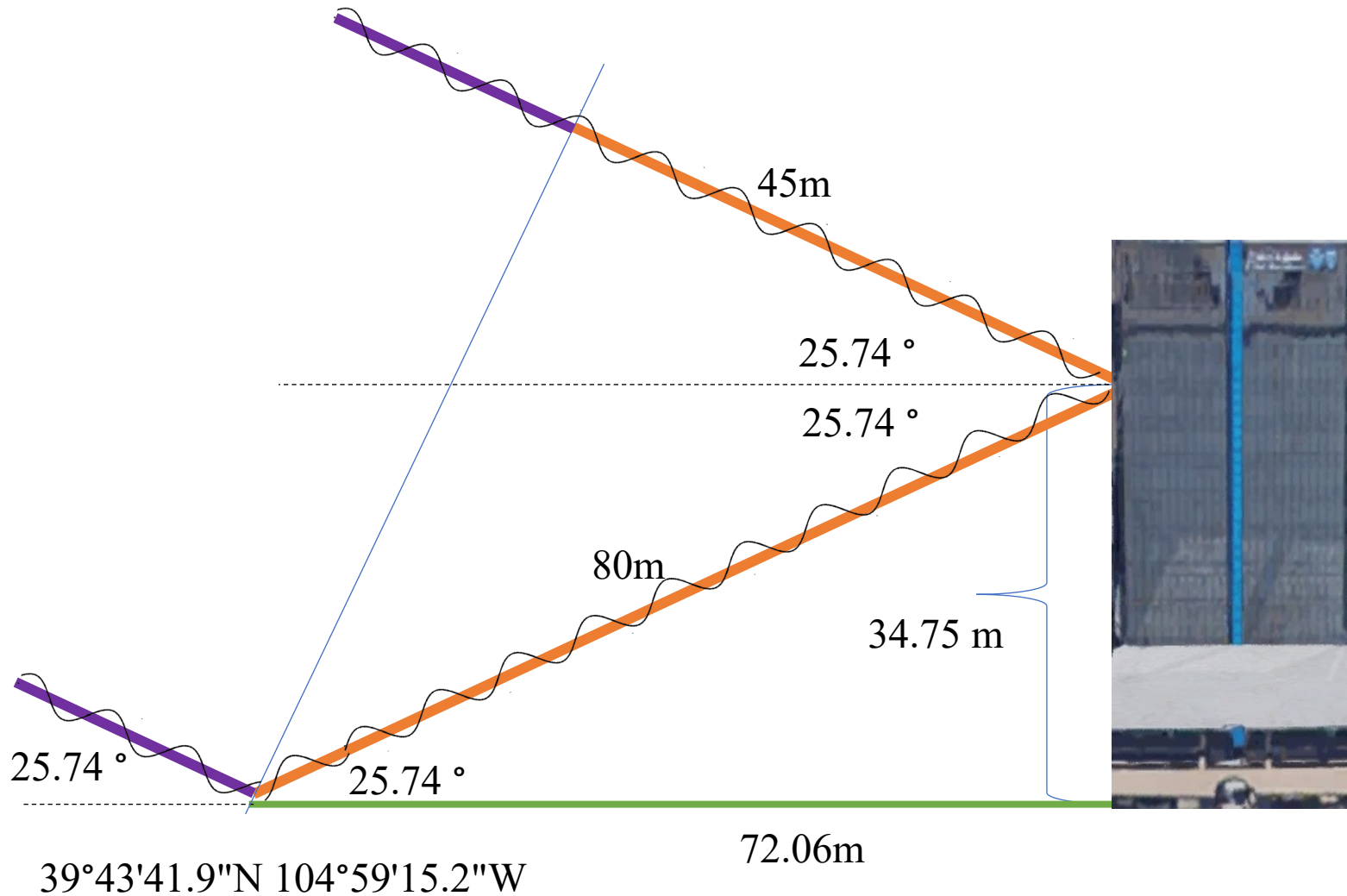
$$SNR_M = \alpha^2 A_D^2 + \alpha 2 A_D^2 \cos \psi$$

Multipath effect on measured signal to noise ratio

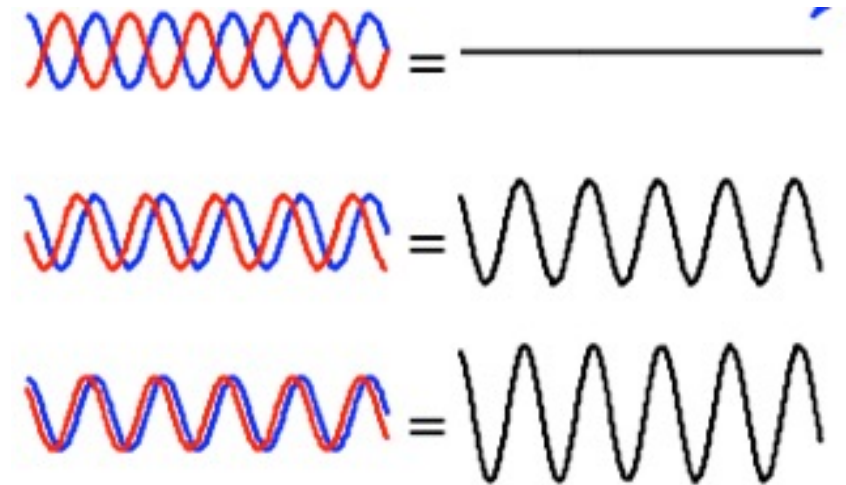
Building Reflection Multipath



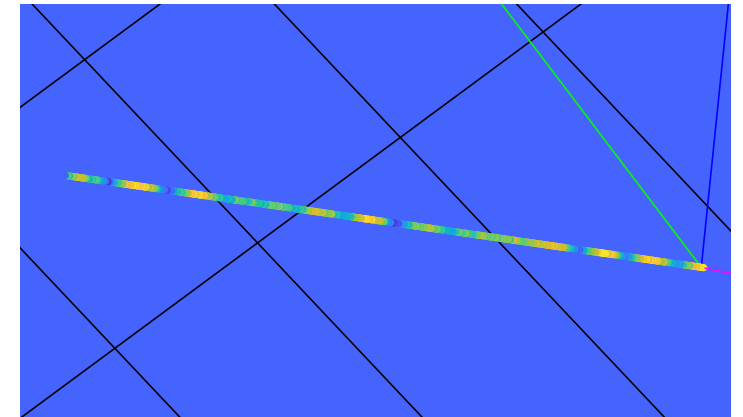
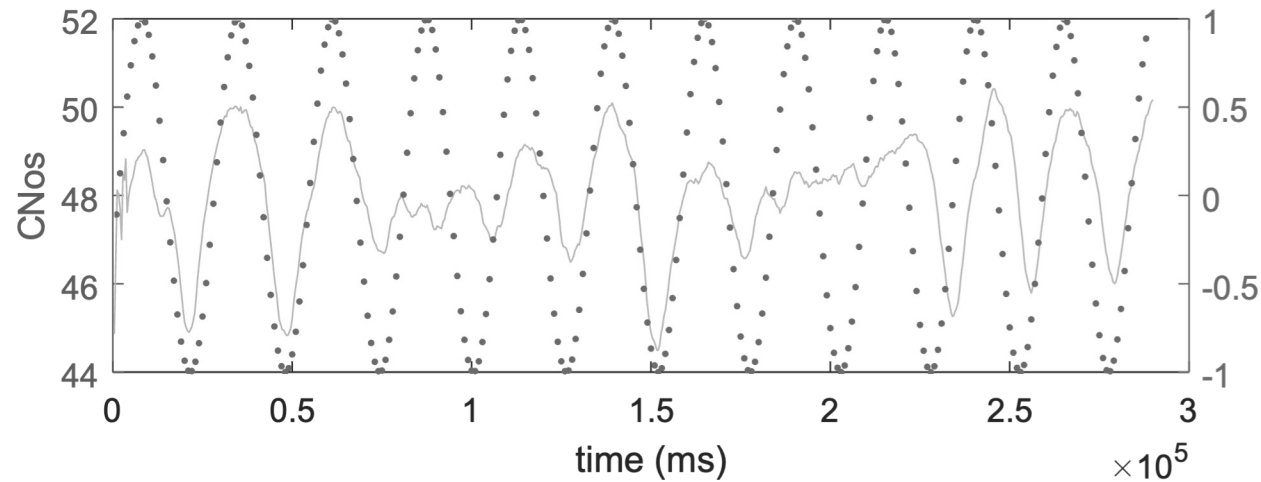
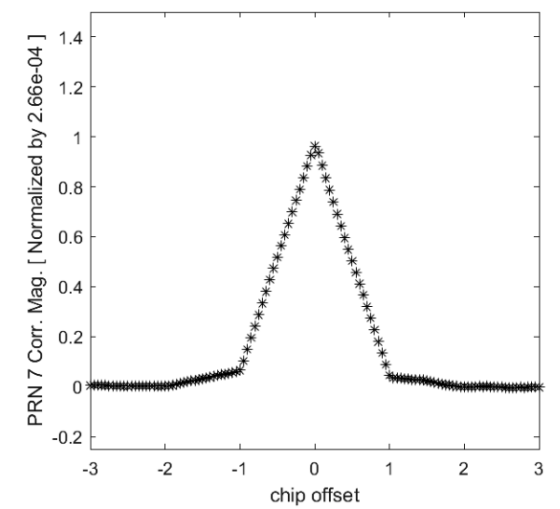
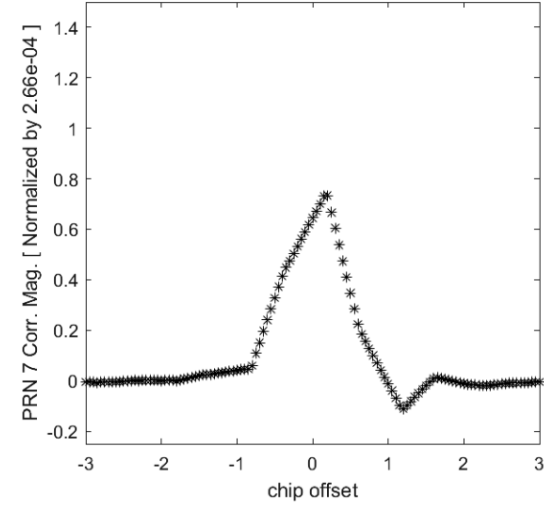
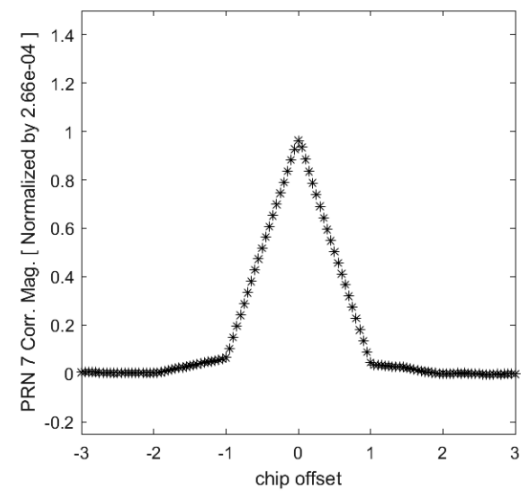
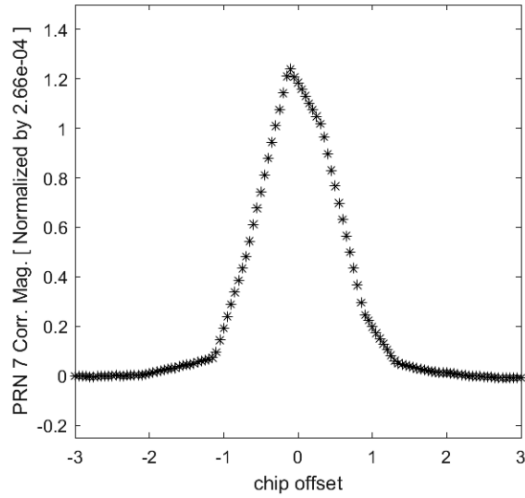
GNSS Multipath Geometry



Intersection Path over ~5min



GNSS Multipath Geometry

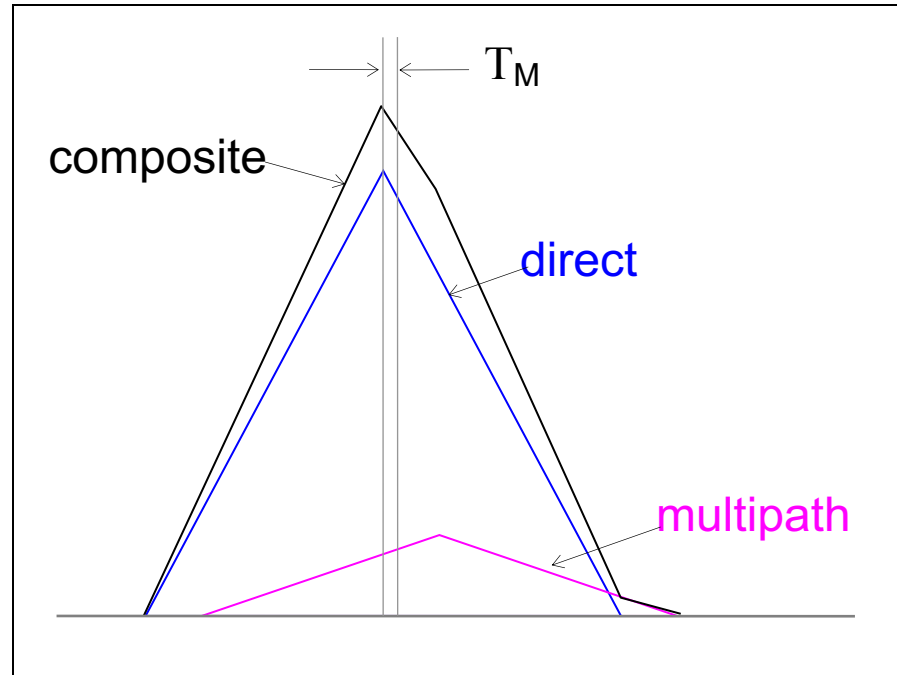


Effect of Multipath on Code Measurement

$$\left[R(\tau_M + T_{E/L}) - R(\tau_M - T_{E/L}) \right] + \alpha \left[R(\tau_M + T_{E/L} - \delta) - R(\tau_M - T_{E/L} - \delta) \right] \cos \psi = 0$$

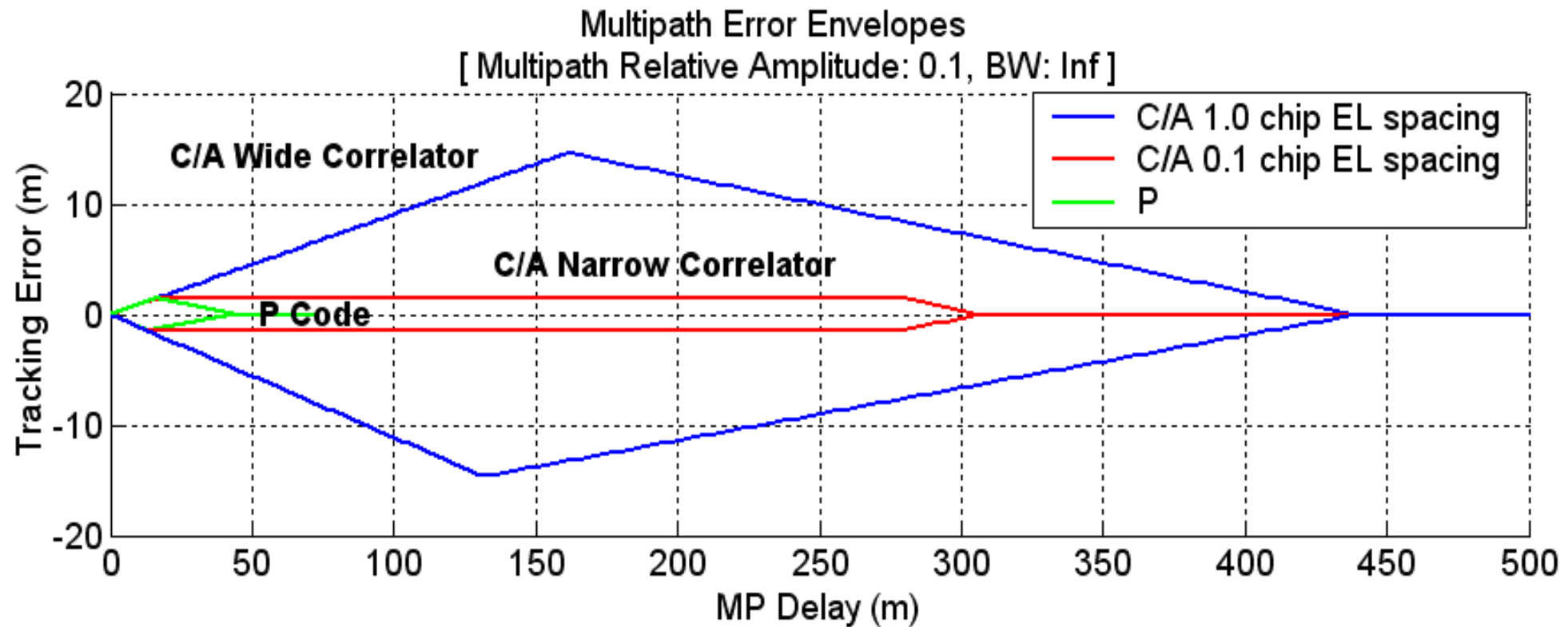
SHORT DELAY Approximation

$$\tau_M = \frac{\alpha \delta \cos \psi}{1 + \alpha \cos \psi}$$



Code Multipath Error Envelope

- Magnitude of reflected signal compared to direct
- Delay of reflected signal relative to direct
- Phase of reflected signal relative to direct
- Receiver design



Multipath Characteristics Recap

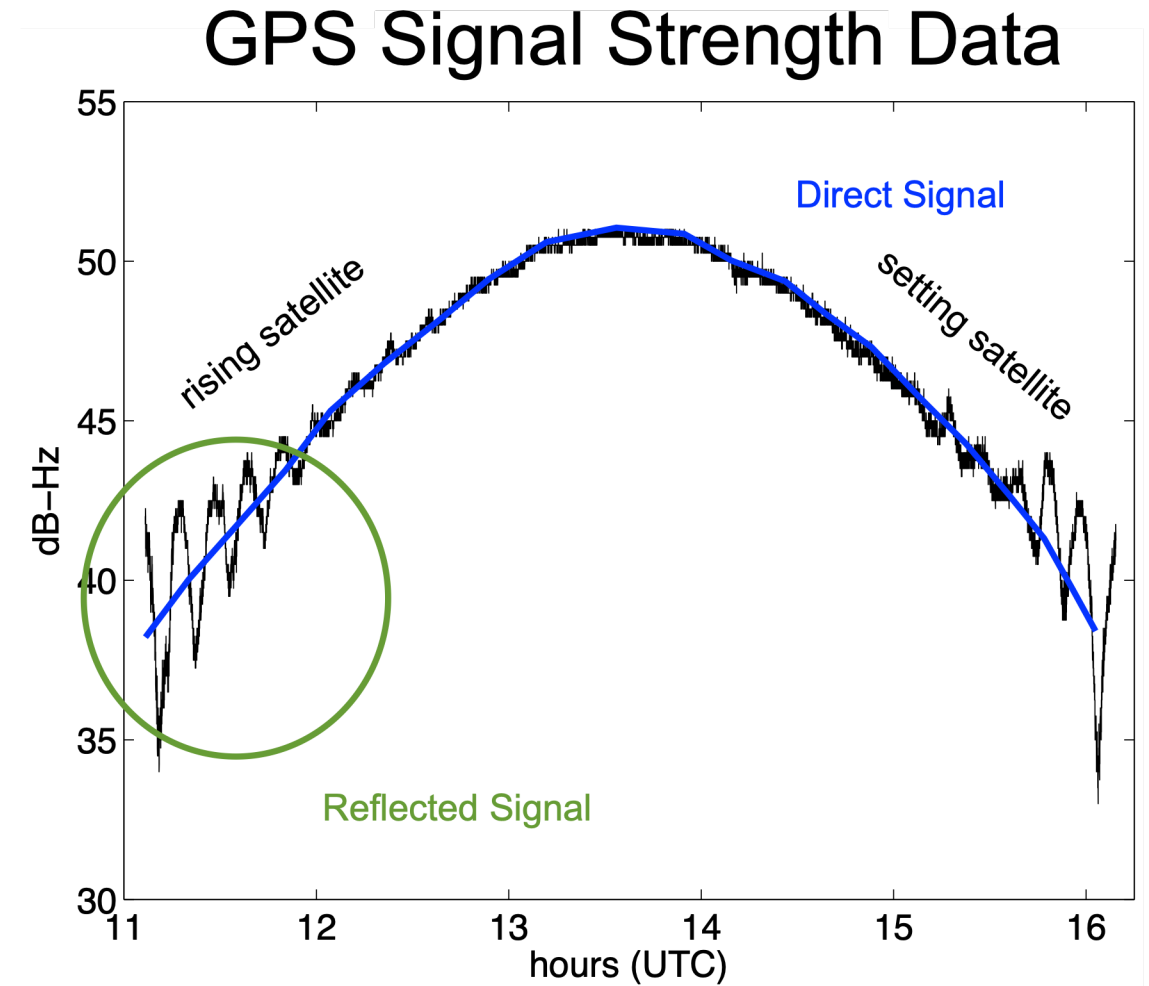
- Multipath “signature” is an oscillation related to the environment near the antenna.
- For a fixed antenna in a static environment, it repeats on a daily basis, shifting in time due to difference between solar day and sidereal day (2 GPS satellite orbits)
- Oscillation frequency is related to reflector distance



CRBT looking to S-SE; Jan 2005.

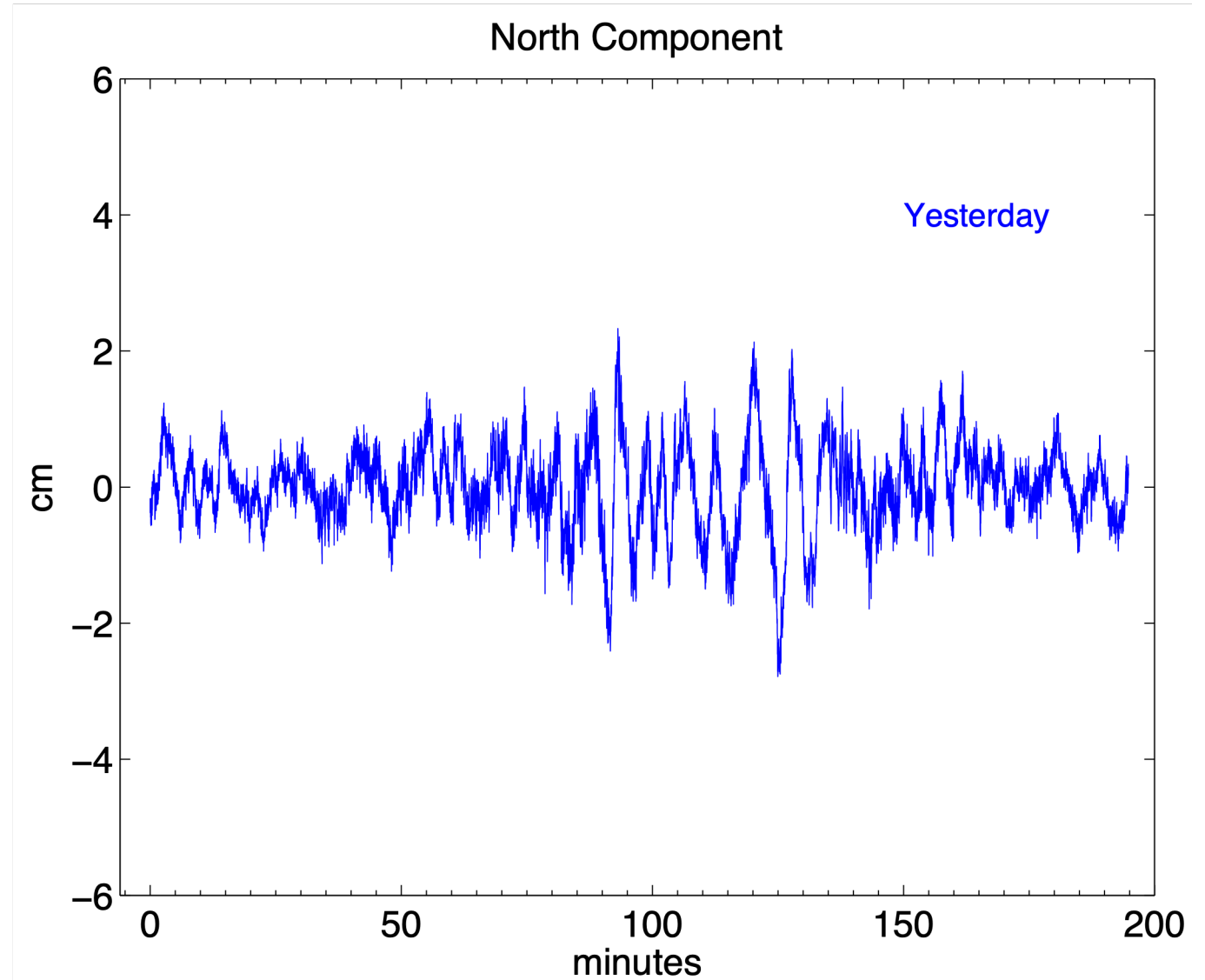


SNR Multipath for an antenna above flat ground

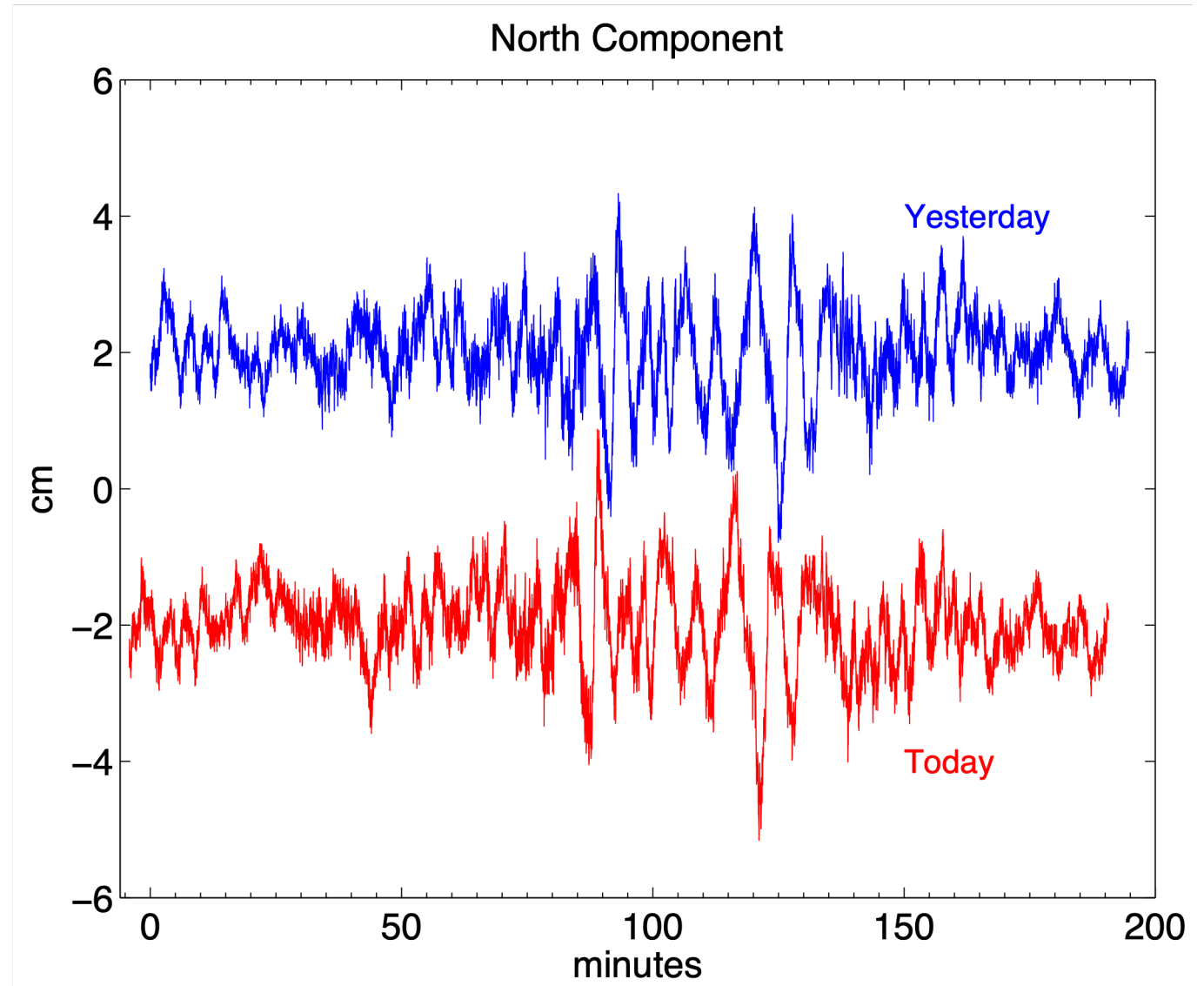


Carrier phase based precise positioning

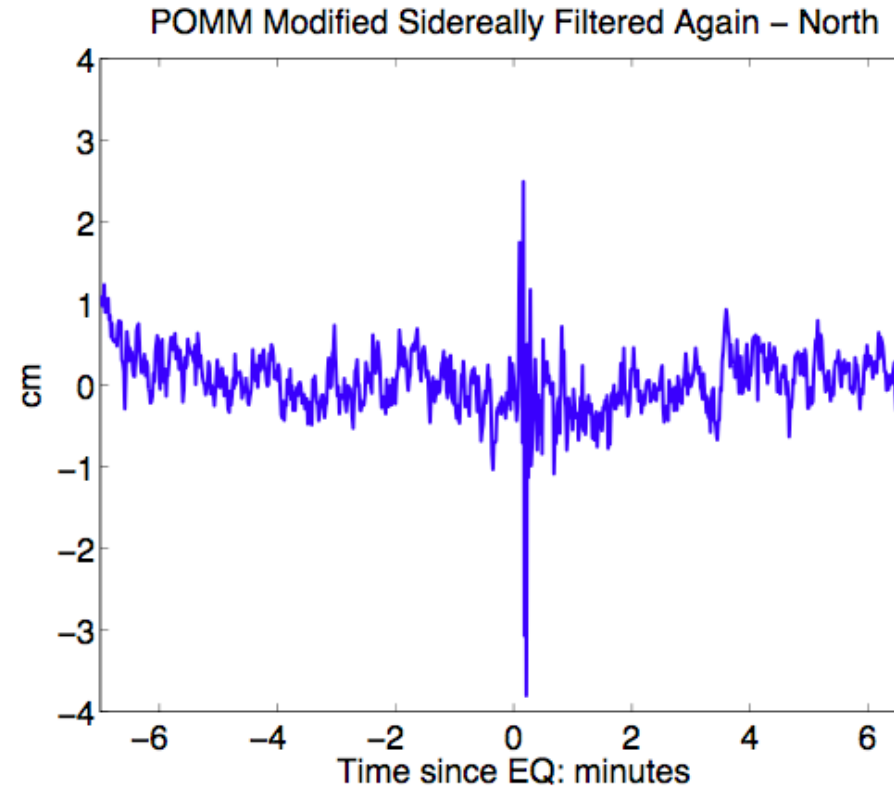
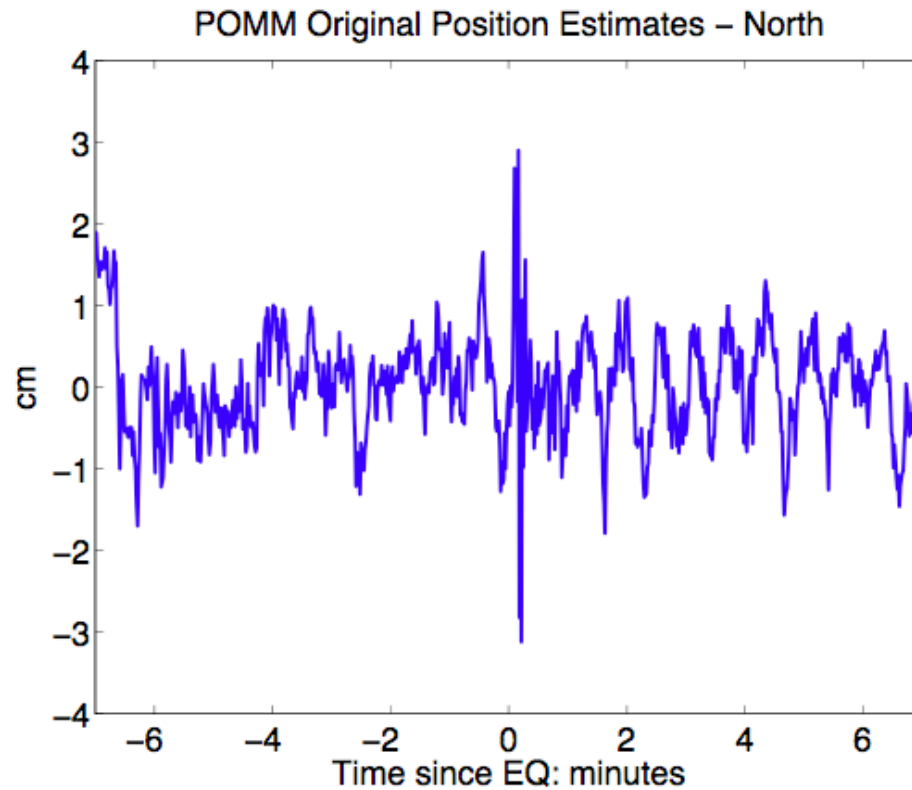
- Example from Prof Larson
- Fixed ground antenna should not be moving this much...



Repeatability



When an earthquake occurs....



Combining observations to see pseudorange multipath

- There is multipath on pseudorange carrier phase, & SNR observables.
- Multipath error on pseudorange is several orders of magnitude larger than multipath on phase.
- How can we isolate multipath?

$$\rho_{L1} = R + c(\delta t_u - \delta t^s) + I_{\rho L1} + T + MP_{\rho L1} + \varepsilon_{\rho L1}$$

$$\lambda_1 \phi_{L1} = R - \lambda_1 N_1 + c(\delta t_u - \delta t^s) + I_{\phi L1} + T + MP_{\phi L1} + \varepsilon_{\phi L1}$$

Code multipath observable

- To evaluate multipath on GPS observables, you need a way to remove all other error sources (orbits, ionosphere, relativity troposphere, clocks).

- With dual frequency observations compute MP1 or MP2:

$$MP_1 = \rho_1 - \frac{f_1^2 + f_2^2}{f_1^2 - f_2^2} \phi_1 \lambda_1 + \frac{2f_2^2}{f_1^2 - f_2^2} \phi_2 \lambda_2 \approx \tau_M + C_0 + \varepsilon_{\rho 1}$$

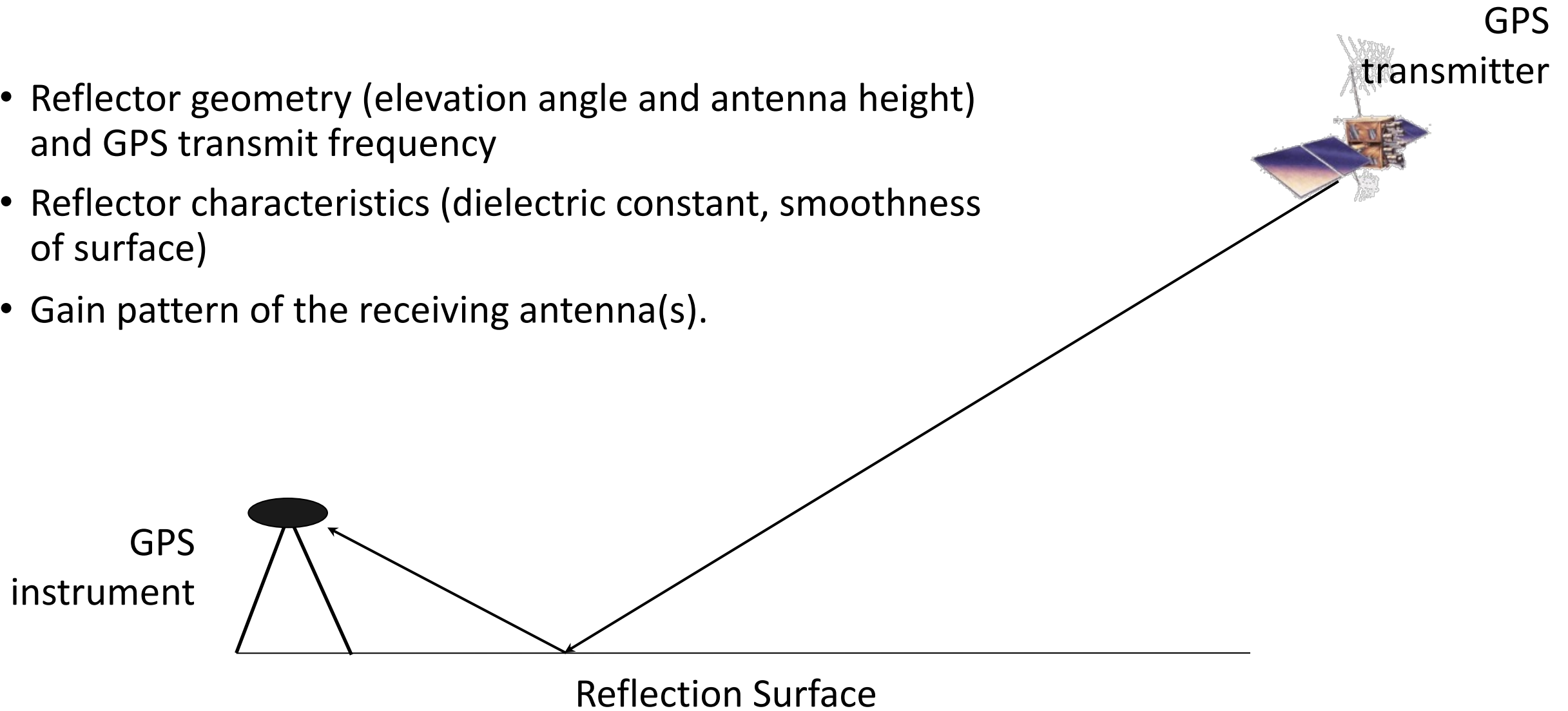
- With only single frequency observations Code minus carrier (CMC) is used to approximate the pseudorange multipath (ignoring contributions of carrier multipath and noise)

$$CMC_1 = \rho_1 - \phi_1 \lambda_1 \approx MP_1 + 2I_1 + C_0 + \varepsilon_{\rho 1}$$

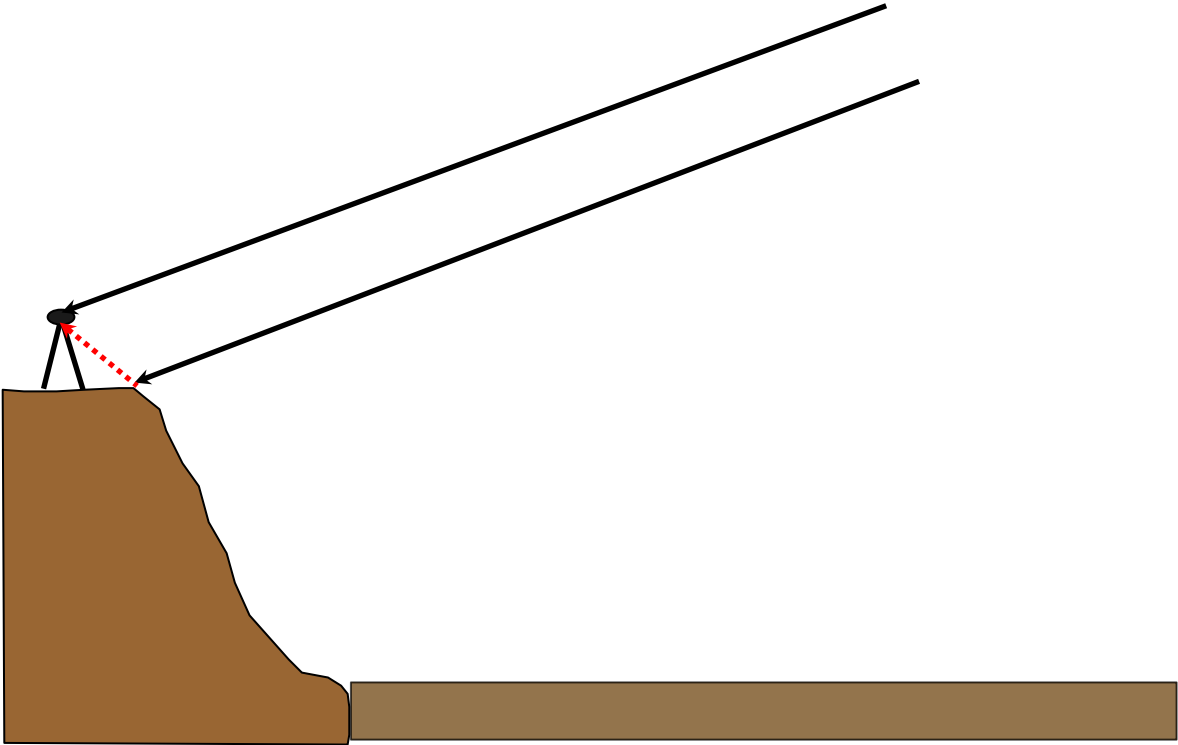


General Problem

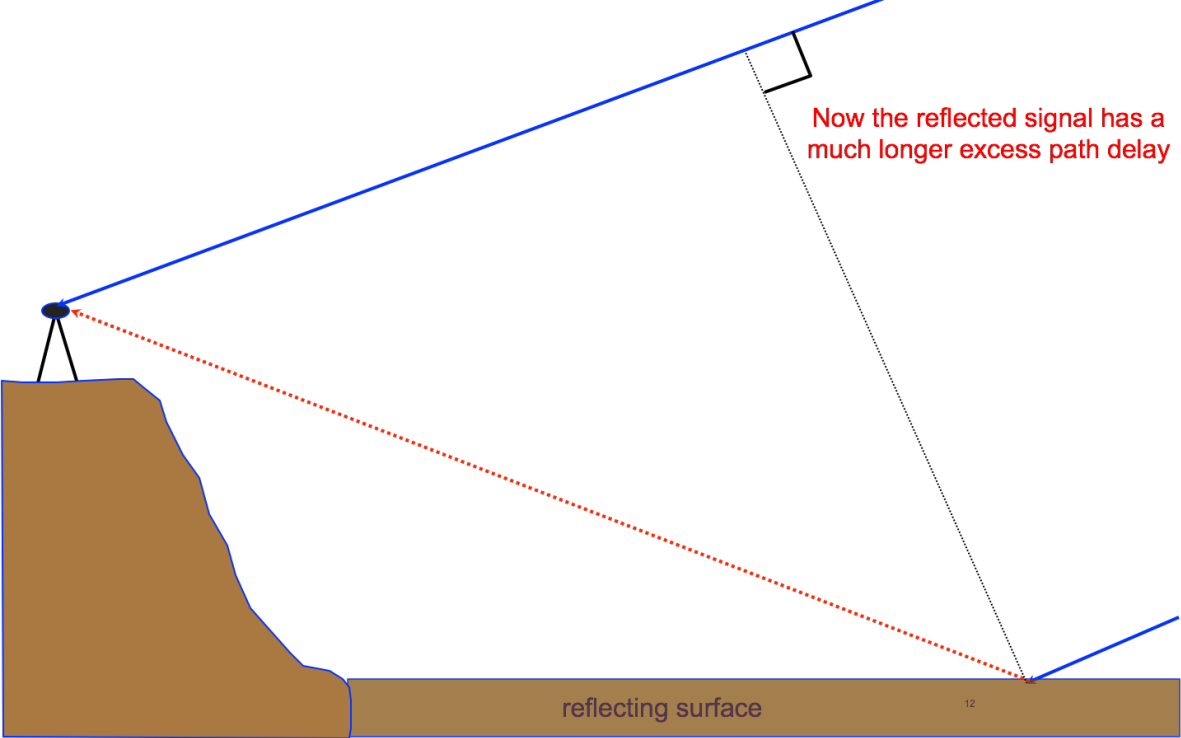
- Reflector geometry (elevation angle and antenna height) and GPS transmit frequency
- Reflector characteristics (dielectric constant, smoothness of surface)
- Gain pattern of the receiving antenna(s).



Close Reflector



Far Reflector



Effect of path delay

Start with excess path length

$$\delta = 2H \sin e$$

Phase delay of the reflected signal:

$$\psi = \frac{2\pi}{\lambda} 2H \sin e$$

Rate of change:

$$\frac{d\psi}{dt} = 4\pi \frac{H}{\lambda} \dot{e} \cos e$$

Change of variable
 $x = \sin e, \frac{d\psi}{dx} = 4\pi \frac{H}{\lambda} = 2\pi \frac{2H}{\lambda}$

For ground reflection:

H – Height of receiver above ground
e – elevation angle of satellite

For building reflection:

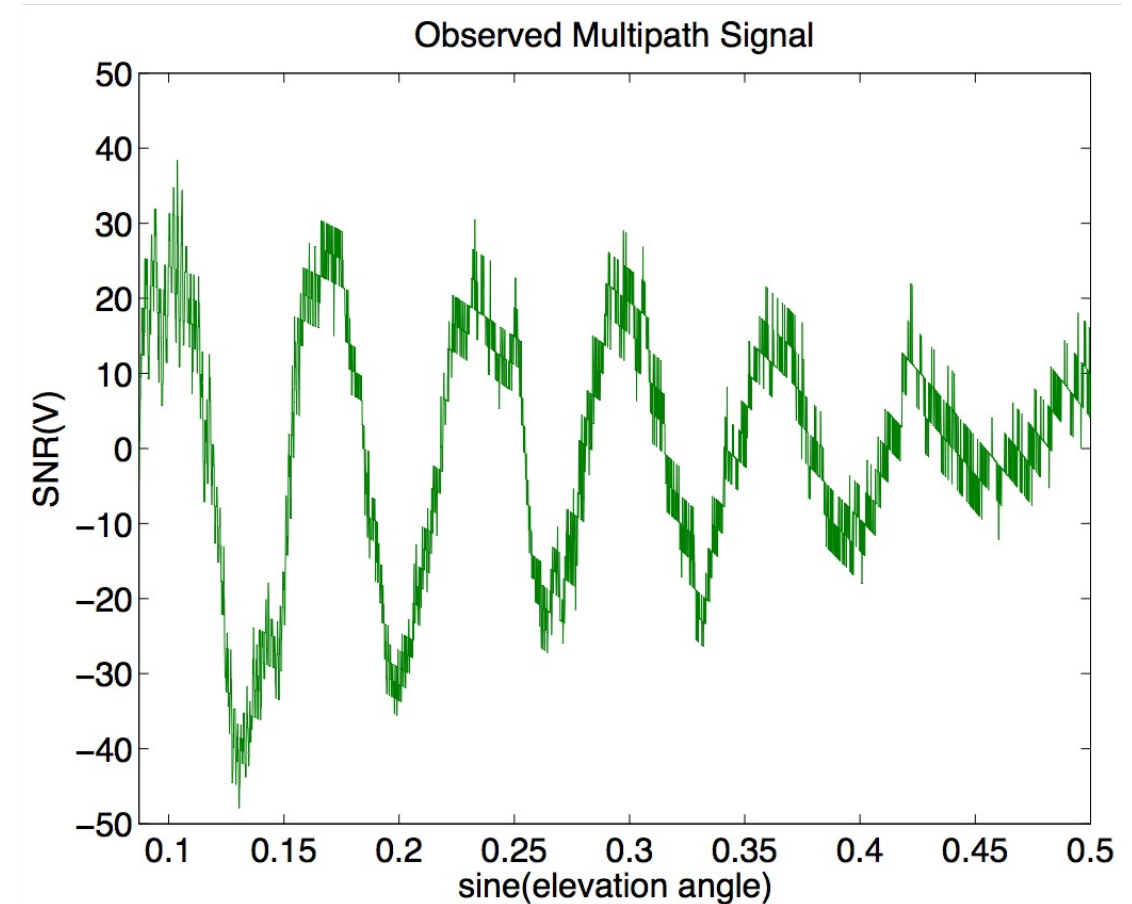
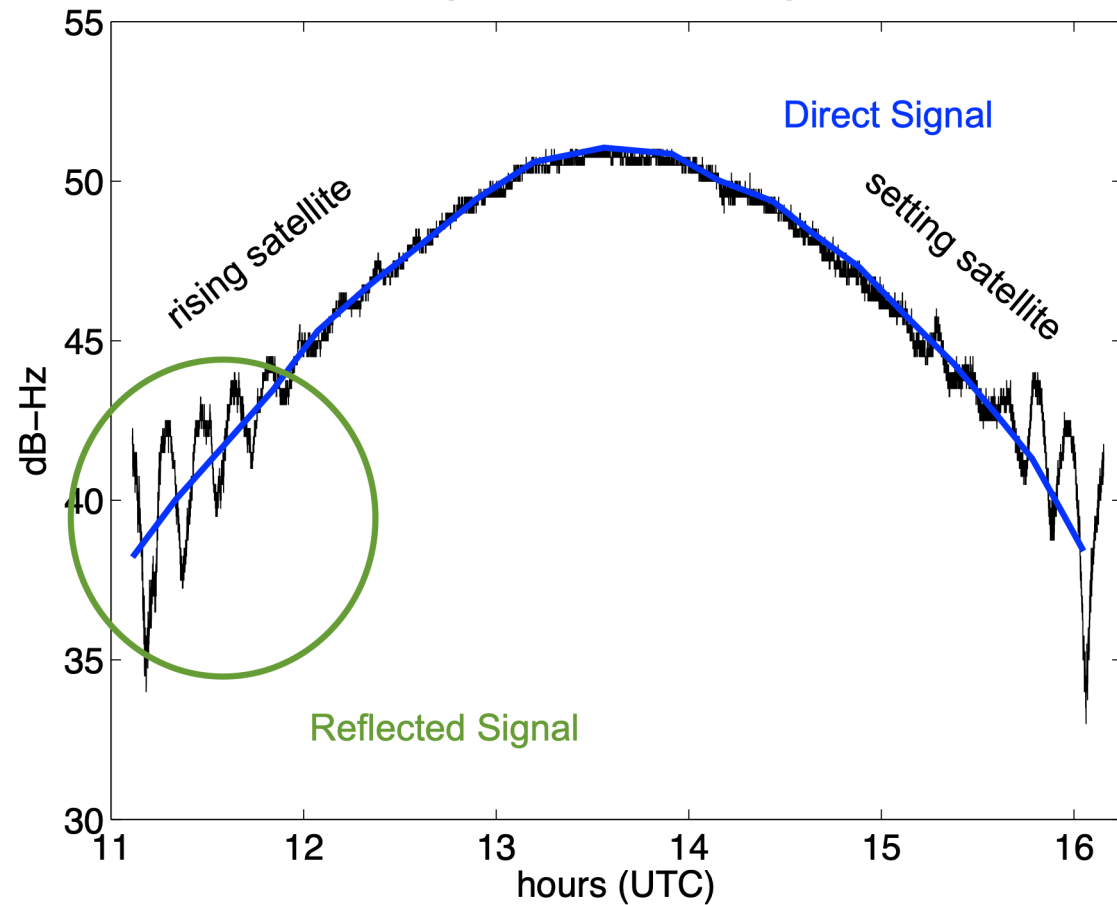
H – Distance of receiver from building
e – ($\pi/2$ - elevation angle of satellite)



GNSS-IR Altimetry

$$SNR = A \cos(2\pi f \sin e + \phi)$$

$$f = \frac{2H}{\lambda}$$



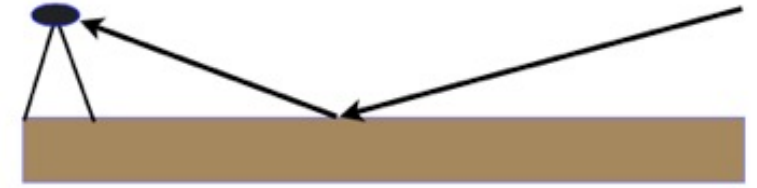
From Prof. Kristine Larson

GNSS-IR Soil Moisture

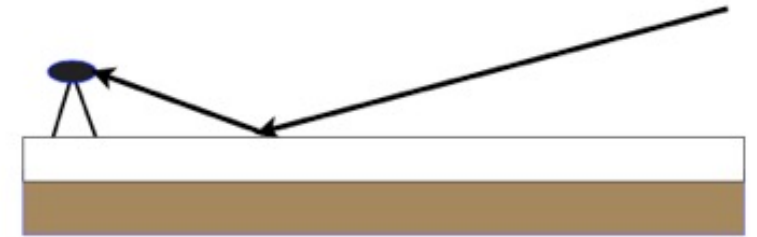
$$SNR = A \cos(2\pi f \sin \theta + \phi)$$

$$f = \frac{2H}{\lambda}$$

the reflections off bare soil produce this SNR curve



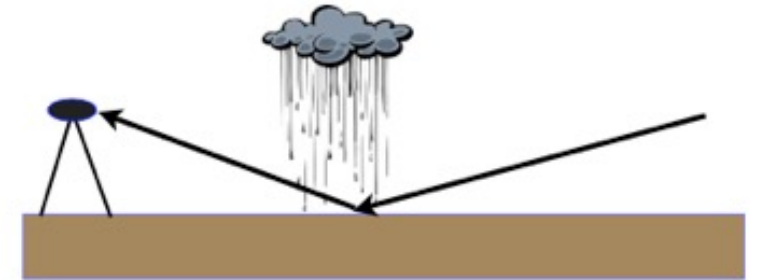
add a snow layer



add vegetation



make the soil wet



From Prof. Kristine Larson



“Accidental tide gauge”

The Accidental Tide Gauge: A GPS Reflection Case Study From Kachemak Bay, Alaska

By Kristine M. Larson,
Richard D. Ray, Felipe G.
Nievinski, and Jeffrey T.
Freymueller

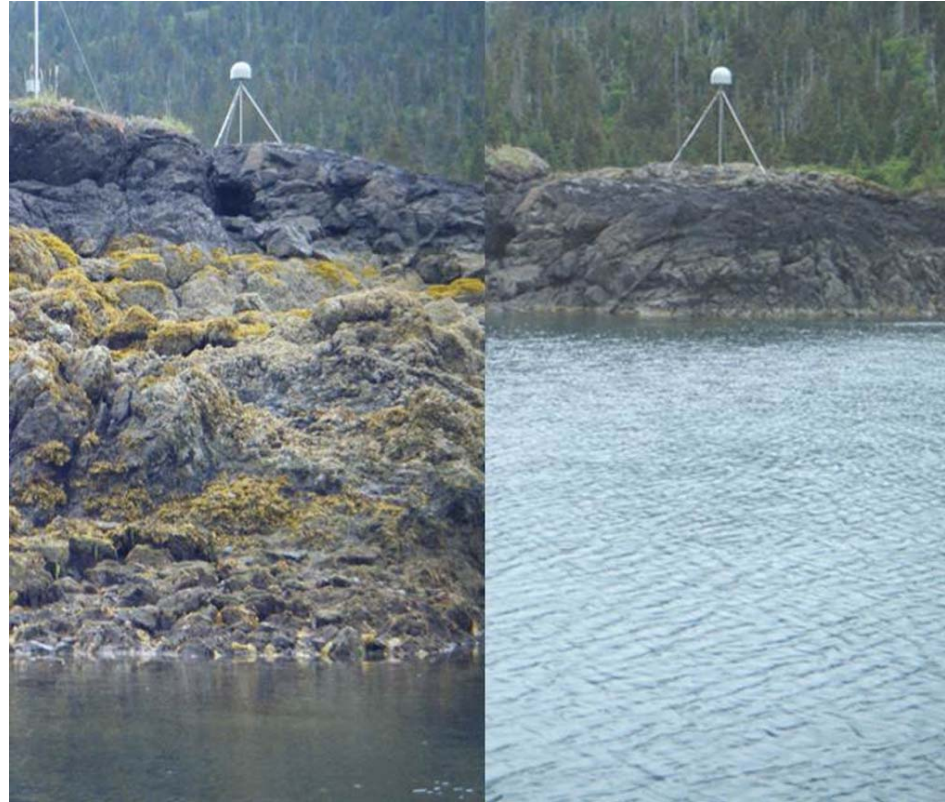


Fig. 1. Photograph of PBAY GPS site at low and high tides.

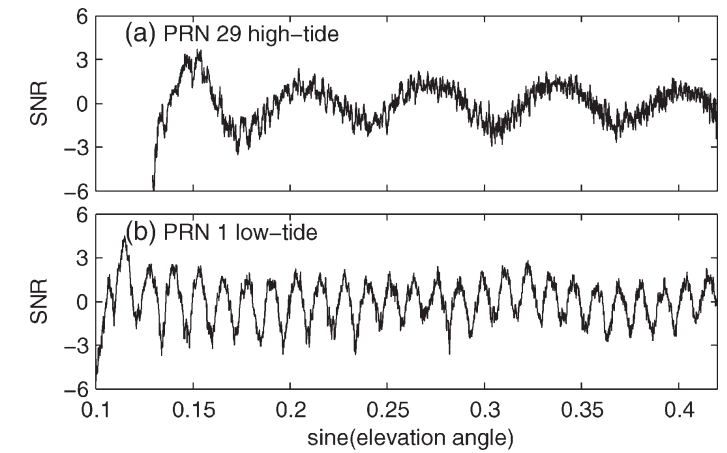


Fig. 2. GPS SNR data (converted from dB-Hz to volts/volts and detrended) for two GPS satellite tracks at high and low tides.

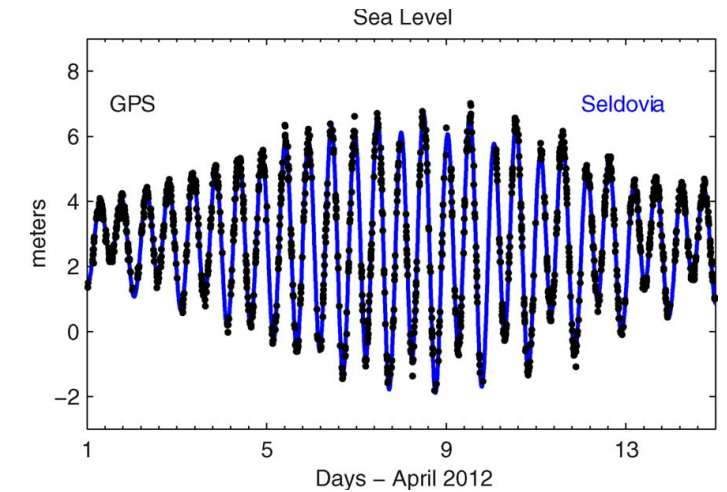
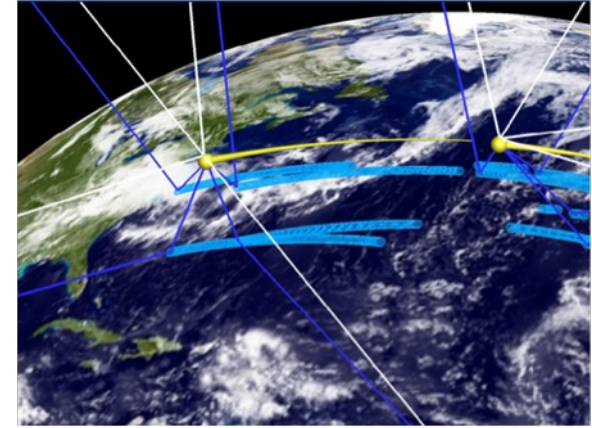


Fig. 3. Time series of sea-level estimates from (circles) GPS and (line) tide-gauge data for Seldovia.

https://www.kristinelarson.net/wp-content/uploads/2015/10/LarsonIEEE_2013.pdf

Using multipath to observe Earth surfaces



Selected References

- Misra & Enge Textbook Ch 5
- Position, Navigation, and Timing Technologies in the 21st Century, Ed by Y.T.J. Morton, et al., Ch 22
Robust Positioning in the Presence of Multipath and NLOS GNSS Signals, Gary A. McGraw, Paul D. Groves, and Benjamin W. Ashman.
- Springer Handbook of Global Navigation Satellite Systems, 2017
Ch. 15 Multipath – Michael S. Braasch
- Weiss, Jan P., P. Axelrad, S. Anderson, “A GNSS Code Multipath Model for Semi-Urban, Aircraft, and Ship Environments,” *NAVIGATION*, Vol. 54, No.4, p. 294-307, 2007.
- Larson, K. M., E. E. Small, E. Gutmann, A. Bilich, P. Axelrad, J. Braun, “Using GPS multipath to measure soil moisture fluctuations: initial results,” *GPS Solutions*, Vol 12 (3), p. 173-177, DOI:10.1007/s10291-007-0076-6, July 2008.
- Axelrad, P., K. Larson, B. Jones, “Use of the Correct Satellite Repeat Period to Characterize and Reduce Site Specific Multipath Errors,” *ION GNSS-2005*, Long Beach, CA, p. 2638- 2648, September 13-16, 2005.

Acknowledgements

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This lecture includes significant contributions by Dr. Kirsten Strandjord.

Next lecture: Measurement error summary & HW discussion